

Rejuvenating aged osteoprogenitors for bone repair

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Aging reduces tissue regeneration capacity, posing challenges for an aging population. In this **fundamental** study, Reeves et al. show that by combining Wnt-mediated osteoprogenitor expansion (using a special bandage) with intermittent fasting, bone healing can be restored in aged animals. By employing rigorous histological, transcriptomic, and imaging analyses in a clinically relevant model, the authors provide **compelling** evidence supporting the conclusions. The therapeutic approach presented in this study shows promise for rejuvenating tissue repair, not only in bones but potentially across other tissues.

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Abstract

Aging is marked by a decline in tissue regeneration, posing significant challenges to an increasingly older population. Here, we investigate age-related impairments in calvarial bone healing and introduce a novel two- part rejuvenation strategy to restore youthful repair. We demonstrate that aging negatively impacts the calvarial bone structure and its osteogenic tissues, diminishing osteoprogenitor number and function and severely impairing bone formation. Notably, increasing osteogenic cell numbers locally fails to rescue repair in aged mice, identifying the presence of intrinsic cellular deficits. Our strategy combines Wnt-mediated osteoprogenitor expansion with intermittent fasting, which leads to a striking restoration of youthful levels of bone healing. We find that intermittent fasting improves osteoprogenitor function, benefits that can be recapitulated by modulating NAD⁺ dependent pathways or the gut microbiota, underscoring the multifaceted nature of this intervention. Mechanistically, we identify mitochondrial dysfunction as a key component in age- related decline in osteoprogenitor function and show that both cyclical nutrient deprivation and Nicotinamide mononucleotide rejuvenate mitochondrial health, enhancing osteogenesis. These findings offer a promising therapeutic avenue for restoring youthful bone repair in aged individuals, with potential implications for rejuvenating other tissues.

Introduction

Flat bones like the sternum, scapula, pelvic bones, and skull bones play crucial roles in the body, including protecting organs and anchoring muscles. Their thin structure makes them prone to fractures (Huelke and Compton, 1983 [↗](#)), leading to pain, disability, and complications like internal bleeding or organ damage (Ren et al., 2020 [↗](#)). As individuals age, flat bones become more brittle, increasing the risk of fractures (Beedham et al., 2019 [↗](#)). Skull fractures, common in head injuries at any age (Carson, 2009 [↗](#)), carry a higher mortality rate even when considering trauma severity (Tsai et al., 2022 [↗](#)). Aging affects the shape and fragility of calvarial bones (Cotofana et al., 2018 [↗](#)) (Torimitsu et al., 2014 [↗](#)), but the impact on osteogenic tissues supporting bone repair, such as the calvarial periosteum and suture mesenchyme remains elusive.

In the calvaria, bones form and repair through intramembranous ossification (van den Bos et al., 2008 [↗](#)). Osteoblasts which deposit pre-mineralized bone tissue (Bianco and Robey, 2015 [↗](#)) arise from CD90⁺ osteoprogenitor populations (Chan et al., 2015 [↗](#), 2013 [↗](#); Kim et al., 2016 [↗](#)), which are derived from heterogeneous populations of skeletal stem cells (SSCs) (Chan et al., 2015 [↗](#), 2013; Kim et al., 2016 [↗](#)). Osteoprogenitors reside in soft tissues alongside endothelial and other cell types that interface with mineralized bone tissue. The two main described osteoprogenitor niches are the periosteum, which overlays all flat bones (Debnath et al., 2018 [↗](#)), and the suture mesenchyme, only present in the skull, between calvarial bones (Wilk et al., 2017 [↗](#)). Localized Wnt ligands (Wnts) play crucial roles in regulating osteogenesis, influencing stem cells, progenitors, and mature osteoblasts (Garcin and Habib, 2017 [↗](#); Liu et al., 2009 [↗](#); Lowndes et al., 2016 [↗](#); Matsushita et al., 2020 [↗](#); Okuchi et al., 2021 [↗](#)). We recently showed that the Wnt3a-bandage, a transplantable polymer that harbors covalently immobilized Wnt3a proteins, combined with an engineered human osteogenic tissue, can promote bone repair in adult immunodeficient mice (Okuchi et al., 2021 [↗](#)).

In this study, using a clinically relevant immunocompetent mouse model, we examined how aspects of the periosteum and suture mesenchyme change with age, how they respond to injury, and the influence of localized Wnts throughout aging.

Young mice achieve almost complete bone repair of calvarial defects considered ‘critical sized’ in adults, which have limited self-healing ability. We found that age-related changes in bone tissue begin to manifest during adulthood, including increased bone porosity, altered cellular mechanobiology, and declines in mitochondrial activity, vascularity, and the proportion and function of CD90⁺ osteoprogenitor cells within the periosteum and suture mesenchyme. While Wnt bandage increases CD90⁺ cells at the defect site regardless of age, this increase only sufficiently promotes repair comparable to young animals in adult mice, not aged mice. Our study characterized a more pronounced decline in osteogenic compartments in aged mice, encompassing reduced CD90⁺ cell proportion, function, and mitochondrial bioenergetics, alongside other contributing factors. This multifaceted decline leads to severe age-related bone repair deficiency, only partially alleviated by the Wnt3a-bandage, underscoring the need for combinatorial approaches.

To counteract age-related decline in bone repair, we combined localized Wnt3a stimulation with systemic metabolic interventions. In aged mice with critical-size defects, alternating-day intermittent fasting restored osteogenic compartment health, improving CD90⁺ cell function as well as bone remodelling and significantly enhanced bone repair to levels comparable to younger animals. This effect was recapitulated by even shorter-term supplementation of the gut bacteria *Akkermansia muciniphila* or nicotinamide mononucleotide (NMN – an NAD⁺ precursor) in aged mice, highlighting the multifaceted mechanisms of action of intermittent fasting. We further

identify mitochondrial dysfunction as a key component in age-related decline in osteoprogenitor function and show that both cyclical nutrient deprivation or NMN rejuvenate mitochondrial health and enhance osteogenesis *in vitro*.

In summary, our findings highlight that aging tissues retain the potential for health improvements through interventions. Our data reveals an intriguing avenue of research where metabolic and microbiome interventions can be repurposed to treat tissue damage in aged repair-deficient individuals.

Results

Parietal bone structure deteriorates with aging

Initially, we analyzed the changes that occur in calvarial bone of female C57BL6/J mice during aging at the macro-structure. Young mice (6 weeks) are within the period of fastest bone formation and growth, Adult mice (13 weeks) at healthy homeostasis, and Aged mice (> 88 weeks) correspond to declining skeletal health as a model of >70 year old humans (Dutta and Sengupta, 2016). We evaluated the calvaria of these animals using micro-computed tomography (microCT). The proportion of the total bone volume taken up by mineralized bone tissue (Bone Volume/Total Volume, BV/TV) significantly decreased with age (Fig. 1a-b). Cortical bone mineral density (Ct.BMD), the measure of the density of the mineralized bone tissue alone, showed that Aged mice have similar mineral density to Young and Adult mice (nonsignificant differences). The overall thickness from outer surface to inner surface of the parietal bone significantly increased with age (Supplementary Fig. 1a-d).

As parietal bone formation and maintenance is regulated by the cells within the periosteum and suture mesenchyme (Debnath et al., 2018; Li et al., 2021; Maruyama et al., 2016), we next defined the changes that occur within these compartments during aging.

Age-related changes in vascularization, mechanobiology, and osteoprogenitor proportions in osteogenic compartments

We aimed to characterize blood vessels, nuclear and cytoskeletal morphology, and the proportion of osteoprogenitors within the periosteum and suture mesenchyme. Initially, we established that 97% endothelial cells in the periosteum and the suture mesenchyme (osteogenic compartments) express Endomucin and CD31 (Supplementary Figure 1e, f). We found that the proportion of Endomucin-positive (Emcn⁺) blood vessels in the periosteum significantly decreases with age (Fig. 1c-d, Supplementary Fig. 1g-j). Further vascular histomorphometry shows that, in the periosteum, blood vessel lumen area is consistent throughout aging, indicating that it is the decrease in the blood vessel number per area of tissue that drives the declining blood vessel density. In addition, we find that the thickness of the periosteum itself also decreases rapidly with age (Supplementary Fig. 1k, l). Taking increased periosteum thickness and density of blood vessels into account, the estimated blood supply (vessel density multiplied by periosteum thickness) of the whole Young periosteum is over 12-fold higher than that of Aged periosteum. In the suture mesenchyme, blood vessels are larger than in the periosteum, and overall, the vascularization remains stable throughout aging (Supplementary Fig. 2a, b).

Vascular phenotypes can result from tissue mechanobiology (Xie et al., 2018). Therefore, we aimed to measure mechanically-driven aspects of cellular morphology in periosteal and suture mesenchymal cells during aging. Nuclear shape is a product of intracellular (Dahl et al., 2008) and extracellular factors, such as the stiffness of the surrounding matrix (Lovett et al., 2013).

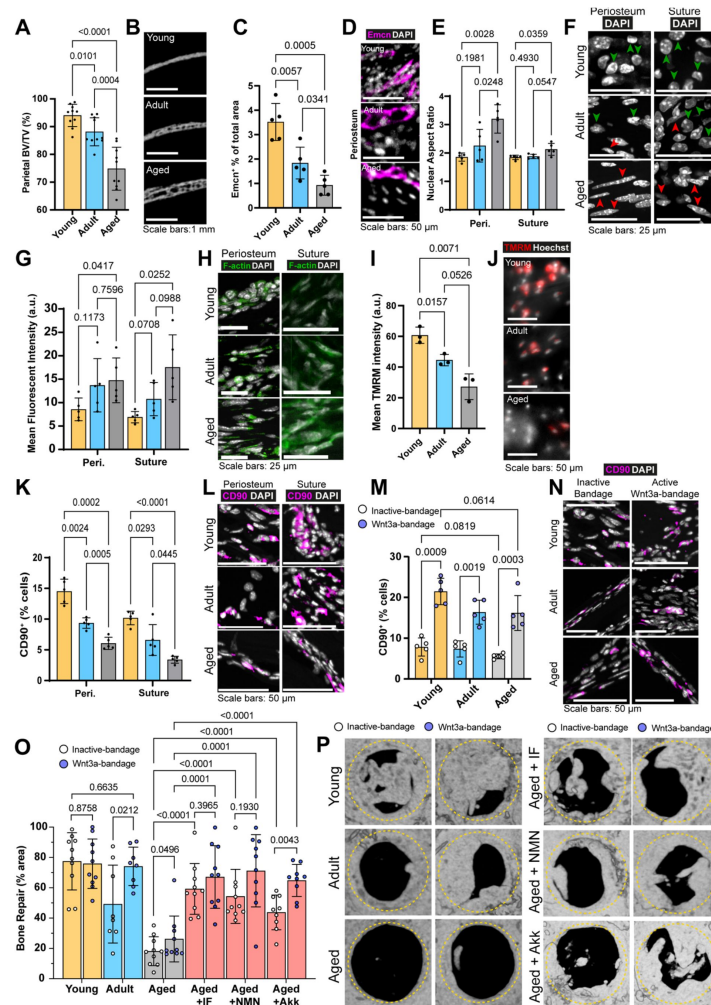


Figure 1

Changes to the calvarial bone, periosteum and suture mesenchyme, and bone repair with aging and interventions (A) Parietal Bone Volume/Total Volume (BV/TV) and (B) representative transverse micro computed tomography (microCT) sections in Young, Adult, and Aged mice. Statistical analysis: Welch's test in $n = 10$ mice per condition. (C) Mean cross sectional area taken up by Endomucin (Emcn)-positive vascular cells and (D) representative immunofluorescence staining of Emcn (magenta) and DAPI (grey) in Young, Adult, and Aged mouse periosteum. Statistical analysis: Welch's tests in $n = 5$ mice per condition. (E) Nuclear aspect ratio (long axis divided by short axis) of cells within the periosteum and suture mesenchyme and (F) representative DAPI-stained nuclei in Young, Adult, and Aged mice. Statistical analysis: Welch's tests in $n = 5$ mice per condition. Green and red arrows highlight rounded and elongated nuclei, respectively. (G) Mean Phalloidin-488 fluorescence intensity and (H) representative images for filamentous actin (F-actin, green) and DAPI (grey) in Young, Adult, and Aged mouse periosteum and suture mesenchyme. Statistical analysis: Welch's tests in $n = 5$ mice per condition. (I) Mean intensity and (J) representative images of live TMRM staining (red) and Hoechst (grey) in Young, Adult, and Aged mouse periosteum *ex vivo*. Statistical analysis: Welch's tests in tissue from $n = 3$ mice per condition. (K) Percentage of CD90⁺ cells within the periosteum and suture mesenchyme, expressed relative to the total cellular number (DAPI⁺) in Young, Adult, and Aged mice. Representative images (N) show CD90 immunostaining (magenta) and DAPI (grey). Statistical analysis: Welch's tests in $n = 5$ mice per condition. (M) Percentage of CD90⁺ cells within calvarial defects treated with Wnt3a-bandages (blue points) or inactive controls (white points), expressed relative to the total cellular number (DAPI⁺) in Young, Adult, and Aged mice. (N) Representative images show CD90 immunostaining (magenta) and DAPI (grey). Statistical analysis: ratio-paired two-tailed t tests in $n = 5$ mice per condition. (O) Percentage of repair and (P) representative top-down microCT images within calvarial defects treated with Wnt3a-bandages (blue points) or inactive controls (white points). Statistical analysis within age groups: ratio-paired two-tailed t tests; between age groups and Aged mice following the IF protocol or NMN or Akk supplementation: Welch's tests; Young, Aged, Aged + IF, Aged + NMN, $n = 10$ mice; Aged + Akk, $n = 9$; Adult, $n = 8$. Complete statistical analysis between aged groups and treatments can be found in **Supplementary Table 1**. The yellow dashed circle highlights the 2 mm diameter defect size.

Morphologically and confirmed by the mean Nuclear Aspect Ratio (NAR, long axis length divided by short axis length) we found significant elongation of nuclei in Aged mice periosteum and to a lesser extent, in the suture mesenchyme (**Fig. 1e-f**).

Nuclear morphology is frequently interlinked with actin cytoskeleton organization, influencing cell migration, proliferation (Ambriz et al., 2018) and response to mechanical force imparted on them through the extracellular matrix (Fletcher and Mullins, 2010; Stricker et al., 2010). We quantified filamentous actin (F-actin) using fluorescent-Phalloidin staining in periosteum and suture mesenchyme (**Fig. 1g-h**, **Supplementary Fig. 2h**). Aging led to a progressive increase of F-actin, peaking at a 2.5-fold rise between young and Aged mice in suture mesenchyme.

Beyond morphological and cytoskeletal changes, we questioned whether aging periosteal cells experience changes to their cellular energy supply. Adenosine triphosphate (ATP) generation, crucial for cell function, relies largely on mitochondrial membrane potential (Zorova et al., 2018). This can be measured by determining the intensity of Tetramethylrhodamine, methyl ester (TMRM), which accumulates in active mitochondria with intact membrane potentials in live cells (Creed and McKenzie, 2019). *Ex vivo* TMRM intensity was significantly higher in freshly-harvested samples of Young mouse periosteum, than in Adult and Aged mice (**Fig. 1i-j**). This finding is unlikely to be an artefact of differential ECM densities interfering with diffusion as performing the same experiment in explanted cells (i.e. without ECM, **Supplementary Fig. 2c, d**) gave a similar trend of results. We also confirmed equal Hoechst diffusion and staining throughout the *ex vivo* tissue experiment (**Supplementary Fig. 2e, f**).

Given these age-related differences in the osteogenic compartments, we aimed to elucidate changes to the CD90⁺ osteoprogenitors (Chan et al., 2015, p. 2; Kim et al., 2016, p. 2), (**Supplementary Fig. 2g**). Using *in situ* immunofluorescence, we found that the proportion of CD90⁺ in the periosteum and the suture mesenchyme significantly decreases with age, with only around one third retained in Aged mice in each compartment (**Fig. 1k-l**).

These tissue and cell-level changes indicate a significant and multifaceted age-related decline in the calvarial osteogenic compartments. The periosteum and the suture mesenchyme also serve as the primary sources of cells that contribute to osteogenesis during calvarial bone repair (Jeffery et al., 2022; Wilk et al., 2017). Hence, we next characterized how calvarial bone repair is affected by age and how the nearby cells respond to the Wnt3a-bandage.

Wnt3a-bandages increase CD90⁺ cells in repair areas of Aged and Adult mice and enhance bone repair

Critical-sized calvarial bone defects, challenging to heal without intervention, serve as models to quantitatively evaluate bone repair. Our prior work demonstrated that applying a Wnt3a-bandage and cellular osteogenic construct to the defect site significantly enhances bone repair in adult immunocompromised mice (Okuchi et al., 2021). Considering the crucial role of the immune system in bone healing (Loi et al., 2016) and the prevalence of bone fractures in immunocompetent individuals, we assessed bone repair in female wildtype (C57BL6/J) mice of varying ages. Mice underwent bilateral critical-sized (2 mm) full-thickness calvarial defect surgery in the parietal bones and were treated with one active Wnt3a-bandage and one chemically-inactivated Wnt3a control bandage. Wnt3a-bandage activity was confirmed *in vitro* (**Supplementary Fig. 3a-c**) before transplantation.

We assessed bone repair by measuring the area percentage of re-mineralized tissue within the defect site using microCT after 10 weeks. Interestingly, Young mice exhibited robust bone healing even without intervention, highlighting the natural regenerative capacity of bone in young mice. (**Fig. 1o-p**, **supplementary table 1**). In adult mice, Wnt3a-bandage treatment significantly enhanced bone repair compared to the control bandage, achieving levels comparable to those

observed in young mice. Aged mice exhibited severely impaired repair with both bandages, though Wnt3a-bandage showed a modest but significant increase. New bone formed under Wnt3a-bandage displayed improved lamellar structure (striations) akin to mature undamaged bone (**Supplementary Fig. 3d**).

To uncover cellular mechanisms behind the improved healing, we examined CD90⁺ osteoprogenitor cells in the defect site. The proportion of CD90⁺ cells was significantly increased by the Wnt3a-bandage across age groups, compared to control bandages (**Fig. 1m-n**). Wnt3a-bandages locally increase CD90⁺ cells, but despite similar numbers across ages, the effect of the Wnt3a-bandage on bone repair in Aged mice was limited. We hypothesized, therefore, that the functionality of CD90⁺ cells, rather than their number alone, may drive age-related differences in bone repair outcomes.

To investigate this, we initially examined the influence of the Wnt-bandage on CD90⁺ cells across various age groups in terms of Wnt/ β -catenin signaling. The proportion of nuclear β -catenin (**Supplementary Fig. 4a-c**), indicative of Wnt/ β -catenin signaling activation (Archbold et al., 2012), was significantly increased in CD90⁺ cells near the active Wnt3a-bandage compared to the inactive bandage. Importantly, the proportion of nuclear β -catenin remained similar regardless of age, suggesting the Aged CD90⁺ have the capacity to activate the pathway similar to younger CD90⁺ cells.

We then explored the effects of the Wnt3a-bandage on CD90⁺ cell proliferation and migration near the bandage. Our results demonstrate that the proportion of CD90⁺ stained with the proliferation marker Ki67 or MMP2, a Type IV collagenase enzyme linked to migration (Ries et al., 2007) and ECM remodeling in bones (Zhong-Sheng et al., 2022), appeared increased in CD90⁺ near Wnt3a-bandage. While this increase was not statistically significant compared to inactive control bandage groups in Aged mice, it was statistically significant in Adult mice (**Supplementary Fig. 4b-e**).

Considering that CD90⁺ cells inside the defect are ultimately derived from cells that are harbored in the nearby periosteum and suture mesenchyme, we next analyzed the transcriptomes of CD90⁺ to glean insights into their functional differences during aging.

Transcriptional shifts occur in calvarial CD90⁺ cells with aging

Single-cell RNA sequencing (scRNAseq) was performed on live cells immediately following dissection and tissue digestion from the periosteum and suture mesenchyme in Young, Adult, and Aged mice, with data processing to deplete immune (CD45⁺) cells. CD90⁺ subpopulations of cells (cluster 0, 1 and 3 in the periosteum; cluster 10 in the suture mesenchyme) were identified within both tissues (**Supplementary Fig. 5a-h**), with high transcriptomic similarity to cells expressing Prrx1, Cathepsin K (Ctsk) and Ddr2 (Bok et al., 2023; Debnath et al., 2018) (**Supplementary Fig. 6**), markers of periosteal and suture SSCs (Duchamp de Lageneste et al., 2018; Wilk et al., 2017). We also found that the CD90⁺ population includes minor subsets of cells that express other SSC/progenitor markers including CD200/CD51 (Cd200/Itgav) (Menon et al., 2021), Gli1 (Zhao et al., 2015), Axin2 (Maruyama et al., 2016) and α -SMA (Acta2) (Ortinou et al., 2019). Therefore, based on functional studies (Kim et al., 2016; Nusspaumer et al., 2017) and our transcriptomic analysis, the CD90⁺ population encompasses a heterogeneous group of cells that are capable of osteogenic differentiation. Next, we will describe selected transcriptomic changes in CD90⁺ cells during aging. For Gene ontology number, comparisons and statistics, please see **Supplementary Table 2**.

Comparing Young CD90⁺ cells with Aged CD90⁺ cells in the periosteum (**Supplementary Fig. 7a**), 542 genes were significantly upregulated. Of these, 38 corresponded to Gene Ontology (GO) term 'Cytosolic ribosome', linked to increased 'Collagen-containing extracellular matrix'. Similarly in Young vs Aged suture CD90⁺ cells (**Supplementary Fig. 7b**), 879 significantly upregulated genes

are enriched for ‘Structural constituent of ribosome’ and ‘Extracellular matrix structural constituent’. Young mice at 6 weeks of age undergo significant skeletal growth(Jilka, 2013), hence this data closely corroborates the increase in collagenous matrix deposition (both soft and bone tissue) observed in histological sections of bone defects in Young mice.

Given the effect of the Wnt3a-bandage on the Aged bone repair environment, we next probed the differentially expressed genes for components of the Wnt/ β -catenin pathway. The transcript’s expression of Wnt2 ligand, often implicated in the Wnt/ β -catenin pathway(Goss et al., 2009; Jung et al., 2015), was significantly downregulated with age in suture CD90⁺ cells (Supplementary fig. 7b,d). Further, Wnt co-receptor Lrp6 expression significantly declined with age in periosteal CD90⁺ cells (Supplementary fig. 7a). Collectively, this suggests compromised Wnt signaling in the periosteum and suture mesenchyme during aging.

At the Adult and Aged timepoints, bone is at healthy and declining homeostasis, respectively. We compared Adult and Aged CD90⁺ cells in the periosteum (Supplementary Fig. 7c). Here, the 637 genes significantly upregulated in Adult vs Aged were vastly enriched for GO terms ‘NADH dehydrogenase complex assembly’, ‘NADH dehydrogenase (ubiquinone) activity’ and ‘Oxidative phosphorylation’. Likewise, in the suture mesenchyme (Supplementary Fig. 7d), the 610 genes significantly upregulated in Adult vs Aged CD90⁺ cells were also enriched for ‘Mitochondrial ATP synthesis coupled proton transport’ and ‘Oxidative phosphorylation’. Using the UCell method(Andreatta and Carmona, 2021), we found significantly reduced Complex I gene scoring with age in the suture mesenchyme and the periosteal CD90⁺ cells (Supplementary Fig. 7e-f). These results, coupled with live measurements of mitochondrial membrane potential (Fig. 1i-j), indicate an age-related decline in mitochondrial metabolism and electron transport chain (ETC) complexes within CD90⁺ cells of the whole periosteum and suture mesenchyme. Importantly, we did not observe distinct changes in the signature of Glycolysis or lipid metabolism across age groups in CD90⁺ cells in the osteogenic compartments (Supplementary Fig 7g-j).

Analysis with the CellChat package(Jin et al., 2021) shows inferred signaling heterogeneity within the periosteal CD90⁺ population (Supplementary Fig. 8a). Cells within the smaller CD90⁺ cluster (cluster 3, Supplementary Fig. 5a-d) receive incoming Wnt-signals and also Periostin-signalling, which upregulates Wnt/ β -catenin pathway activity in osteogenesis(Lv et al., 2015; Zhang et al., 2017). This cluster is significantly upregulated for markers of osteogenic differentiation such as Runx2, Mme (Osteonectin/CD10)(Ding et al., 2020; Granéli et al., 2014), Bgn(Ye et al., 2012), Mgp(Zhang et al., 2019), Gas6(Shiozawa et al., 2010), and Igfbp7(Zhang et al., 2018), and is hereby termed the P1 CD90⁺ population. Cells within the larger cluster (clusters 0 & 1) both send and receive Periostin and Wnt-signals, and is significantly upregulated for markers associated with stemness, such as Klf4 and Aldh1a3(Wanandi et al., 2018), and negative regulators of osteogenic differentiation such as Bmp3(Kokabu et al., 2012), and Ccn3(Matsushita et al., 2013). Hereby these cells were termed P2 CD90⁺.

The periosteal P2 CD90⁺ cell population shifts with age between two categories, named ‘Igfbp7^{high}’ and ‘Igfbp7^{low}’ in relation to their differential expression of Igfbp7 (Fig. 2a-c). Igfbp7 is linked to bone anabolism through its dual roles in enhancing Wnt-signaling and exerting metabolic effects(Locatelli and Bianchi, 2014). Young P2 CD90⁺ cells were restricted to the Igfbp7^{high} grouping (92.0%), while Aged CD90⁺ cells were predominantly in the Igfbp7^{low} grouping (66.3%) (Fig. 2a-b). Adult CD90⁺ cells were more evenly split (56.8% Igfbp7^{high}, 43.2% Igfbp7^{low}). Igfbp7^{low} cells were significantly upregulated for 85 genes (Fig. 2c), linked ‘Cell motility’, driven by genes such as Mmp14(Almalki and Agrawal, 2016) and Itgb7(Novoseletskaia et al., 2020). Igfbp7^{high} cells were significantly upregulated for 204 genes linked to ‘Collagen-containing extracellular matrix’, ‘Oxidative phosphorylation’ and ‘Respiratory chain complex’. The enrichment is driven by significantly upregulated genes such as NADH dehydrogenase genes (mt-Nd1 & mt-Nd2), Col1a1, Sparc.

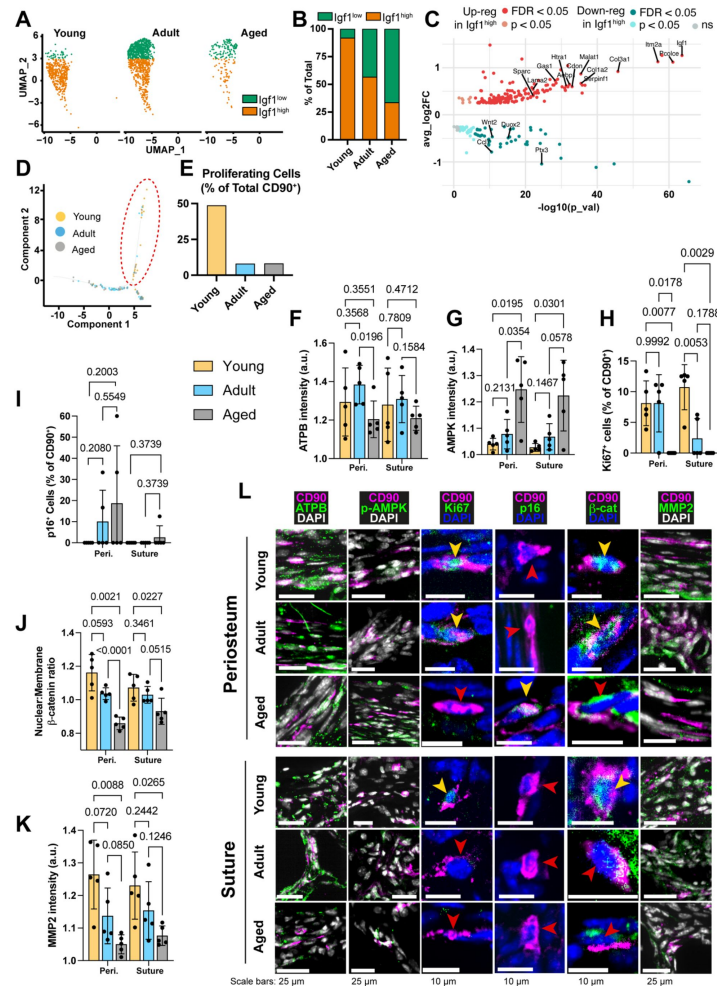


Figure 2

Transcriptomic and protein expression characterization of CD90⁺ cells during aging

(A) UMAP-plot showing the distribution of a subpopulation of CD90⁺ periosteal cells with aging, categorized as Igf1^{high} (orange) and Igf1^{low} (green). (B) Percentage of the total CD90⁺ subpopulation categorized as either Igf1^{high} or Igf1^{low}. Young: Igf1^{high} (n = 425 cells), Igf1^{low} (n = 37); Adult: Igf1^{high} (n = 553), Igf1^{low} (n = 421); Aged: Igf1^{high} (n = 61), Igf1^{low} (n = 120). (C) Volcano plot showing differentially-expressed genes in the comparison of all Igf1^{high} cells (n = 1039) against all Igf1^{low} cells (n = 578). Statistical analysis: Wilcoxon Rank Sum test via the FindMarkers function in the Seurat R package; upregulated (pink: p < 0.05; red: False discovery rate (FDR) -adjusted p < 0.05), downregulated (cyan: p < 0.05; teal: FDR-adjusted p < 0.05), and non-significant (ns) genes are displayed on axes of log fold expression (Igf1^{high} over Igf1^{low}) (y, avg_log2FC) against p transformed as its negative log value. (D) Dimensional reduction plot (Monocle package(Trapnell et al., 2014)) of suture mesenchyme CD90⁺ populations in Young (yellow), Adult (blue), and Aged (grey) mice. (E) Percentage of CD90⁺ categorized as 'proliferating' (inside red dashed ellipse) vs 'non-proliferating' (outside ellipse). Young: proliferating (n = 19 cells), nonproliferating (n = 20); Adult: proliferating (n = 5), non- proliferating (n = 57); Aged: proliferating (n = 5), nonproliferating (n = 56). (F) Normalized intensity staining for ATPB in CD90⁺ cells. (G) Normalized intensity staining for phospho-AMPK in CD90⁺ cells. (H) Proportion of CD90⁺ cells that are Ki67⁺. (I) Proportion of CD90⁺ cells that are p16⁺. (J) Ratio of Nuclear:Membrane β -catenin staining in CD90⁺ cells. (K) Normalized intensity staining for MMP2 in CD90⁺ cells. For all plots (F-K), Young (yellow), Adult (blue), Aged (grey), periosteum (3 left columns), suture mesenchyme (3 right columns), statistical testing: Welch's tests in n = 5 mice per condition. (L) Representative images depicting CD90 (magenta), DAPI (grey/blue as indicated by column), and functional markers (green) as indicated by column. Yellow arrows indicate marker-high/positive cells; red arrows indicate marker-low/negative cells.

Likewise, in the suture mesenchyme, analysis of inferred cellular state and function (**Fig. 2d-e**) revealed a distinct subpopulation of CD90⁺ cells. This group of cells was significantly upregulated for 855 genes linked to ‘Mitotic cell cycle’ and ‘ATP-dependent activity’. In Young mice, this population, termed ‘Proliferating’ cells, represented 48.72% of the CD90⁺ population, whereas it markedly declined to around 8% in both Adult and Aged mice (**Fig. 2d-e**).

When comparing CD90⁺ cell populations between every two age stages, several genes are conserved in all three comparisons (**Supplementary Fig. 8b, c**), including: *Cavin1* (Volonte and Galbiati, 2020), *S100a4* (Wagner et al., 2009), *Vcan* (S. Li et al., 2019), *Mmp14* (Haage et al., 2014), and *Nsg1* (Wiese et al., 2019). Each of these genes has been identified as being involved in senescence or the cellular response to aging tissues. Likewise, in the suture CD90⁺ cells, conserved aging-related genes included senescence-related genes *Pim1* (Yang et al., 2017), *Pdcd4* (Kang et al., 2002); and the senescence-associated secretory profile (SASP) gene *Ccl6* (Gillispie et al., 2021). Further comprehensive senescence signature confirms the significant increase associated with age in CD90⁺ cells (**Supplementary Fig. 8d**). Also conserved were: pro-bone resorption *Mmp9* (Zhu et al., 2020), and adipogenesis-related genes *ApoD* (Provost et al., 1991) and *Slc16a3* (MCT4) (Petersen et al., 2017).

Aging calvarial CD90⁺ cells undergo a decline in energy supply, proliferation, Wnt/ β -catenin, and ECM remodeling capacity

Next, we aimed to investigate the transcriptionally-identified decline in energy supply, proliferation, Wnt/ β -catenin pathway, and migration in CD90⁺ cells at the protein level through immunofluorescent staining of marker proteins.

The catalytic beta subunit of ATP synthase (ATPB), an essential component of the mitochondrial ATP synthesis machinery, serves as a marker for assessing energy-related mitochondrial function (H.-S. Li et al., 2019). Cellular ATPB levels in suture mesenchyme CD90⁺ cells is similar across age groups, but in the periosteal CD90⁺ cells, ATPB level is lower in Aged mice compared to Adults (**Fig. 2f, I**). Phosphorylated AMPK (phospho-AMPK), a sensor of ATP depletion (Gowans and Hardie, 2014), exhibits a negative correlation with energy availability. In both periosteum and suture CD90⁺ cells, mean phospho-AMPK was significantly higher in Aged mice (**Fig. 2g, I**). Taken together, these findings suggest a potential deficiency in oxidative phosphorylation machinery in CD90⁺ cells with age.

Ki67 marks active proliferation, while cyclin-dependent kinase inhibitors p16 is involved in various cellular functions and is linked to senescence and aging in several tissues (Safwan-Zaiter et al., 2022; Stein et al., 1999, p. 199). Calvarial CD90⁺ cells were assessed for nuclear expression of the two markers. Ki67⁺ CD90⁺ cells were observed in the periosteum of both Young and Adult mice but not in Aged mice. In the suture, the proportion of Ki67⁺ CD90⁺ was significantly higher in Young than in Adult mice with none detected in Aged mice (**Fig. 2h, I**). The proportion of p16⁺ CD90⁺ cells appeared to increase with Age (**Fig. 2i, I**). Together, nuclear p16 expression and a lack of Ki67 expression suggests that Aged CD90⁺ cells are not actively proliferating, corroborating inferred findings from transcriptomics.

Next, we assessed the level of nuclear β -catenin, an indicator of Wnt/ β -catenin pathway activation. Aged CD90⁺ cells in both periosteum and suture mesenchyme showed significantly reduced nuclear localization of β -catenin compared to Young (**Fig. 2j, I**). Only Adult CD90⁺ cells in the periosteum had significantly higher nuclear β -catenin when compared to Aged mice.

Finally, in both periosteum and suture mesenchyme, Aged CD90⁺ cells expressed the lowest amounts of MMP2 (**Fig. 2k-l**). While MMP2 is a secreted protein, only intracellular protein was quantified, to enable the analysis to be restricted to MMP2 expressed by CD90⁺ cells alone. It

corroborates the downregulation of genes associated with migration in the transcriptomic analysis.

Overall, our protein-level analysis largely aligns with the transcript-level findings, insinuating that age-related alterations in periosteum and suture mesenchyme, and consequently bone repair, could be considerably impacted by metabolic differences within the CD90-expressing population.

Given that increasing the number of CD90⁺ cells inside the defect alone produced only a minor improvement to Aged bone repair, we developed a systemic approach to target cellular metabolism. We aimed to combine this systemic approach with the use of the Wnt3a-bandage to ultimately facilitate the supply of competent osteogenic cells for bone repair.

Intermittent Fasting improves bone health and bone repair in Aged mice

Intermittent Fasting (IF) can improve metabolic control by various measures (Longo et al., 2021) and bone mineral density in adult animals (Veronese and Reginster, 2019). We produced critical-sized calvarial defects in Aged mice, treated them with Wnt3a-bandages and inactive controls. These mice were then maintained on an alternating-day diet of self-limited feeding and complete fasting, hereby termed the intermittent fasting (IF) diet. Importantly, over the course of a week, mice intake the same amount of food during the IF diet as during the self-limited feeding controls (*ad libitum*, AL) diet (Supplementary Fig. 9a), and is therefore distinct from an overall reduction in calories (i.e., caloric restriction).

Remarkably, Aged mice subjected to the IF diet repaired 59.24% of the defect within 10 weeks under the inactive bandage, higher even than that of Adult mice + inactive bandage (Fig. 10-p). With the addition of the Wnt3a-bandage, bone repair area was increased to 67.10%. This combination approach of the local Wnt3a-bandage and post-fracture IF results in Aged calvarial bone repair rivaling that of Young mice. The IF diet also significantly increased the solid bone proportion (BV/TV) (Fig. 3a-b), making it similar to the much younger Adult bone.

Intermittent Fasting reverses aspects of cellular and tissue aging in the periosteum and suture mesenchyme

Compared to Aged mice, in Aged + IF, Emcn⁺ vascularity in the periosteum was significantly increased (Fig. 3c-d). Additionally, in the suture mesenchyme, actin cytoskeleton staining was markedly decreased (Fig. 3e-f), along with a significant reduction in age-related nuclear elongation both in the suture and periosteum (Fig. 3g-h). The IF diet appeared to elevate the proportion of CD90⁺ cells in the periosteum while in the suture, it significantly increased compared to *ad libitum* (Fig. 3i-j).

To gain a deeper understanding of how IF affects Aged CD90⁺ cells, we conducted further scRNAseq analysis. We maintained Adult and Aged mice on the alternating-day IF diet for 10 weeks before performing scRNAseq on the periosteum and suture mesenchyme.

Comparing CD90⁺ cells of Aged periosteal post IF diet with Aged AL diet controls revealed 98 significantly upregulated genes (Fig. 4a). These were enriched mainly for ‘Response to cytokine’ and similar GO terms. Of the genes upregulated with the IF diet, 10 were individually statistically significant when adjusting for false discovery rate ($q < 0.05$). Ccl11 and Duox2 are involved in osteogenic stem cell migration (Smith et al., 2012; Tyurin-Kuzmin et al., 2016), Tnfaip6 and Ccl7 in immunomodulation (Li et al., n.d.; Rafei et al., 2008), Ptx3 is involved in both processes (Cappuzzello et al., 2016), and Actg1 in cytoskeletal structure (Dugina et al., 2009);

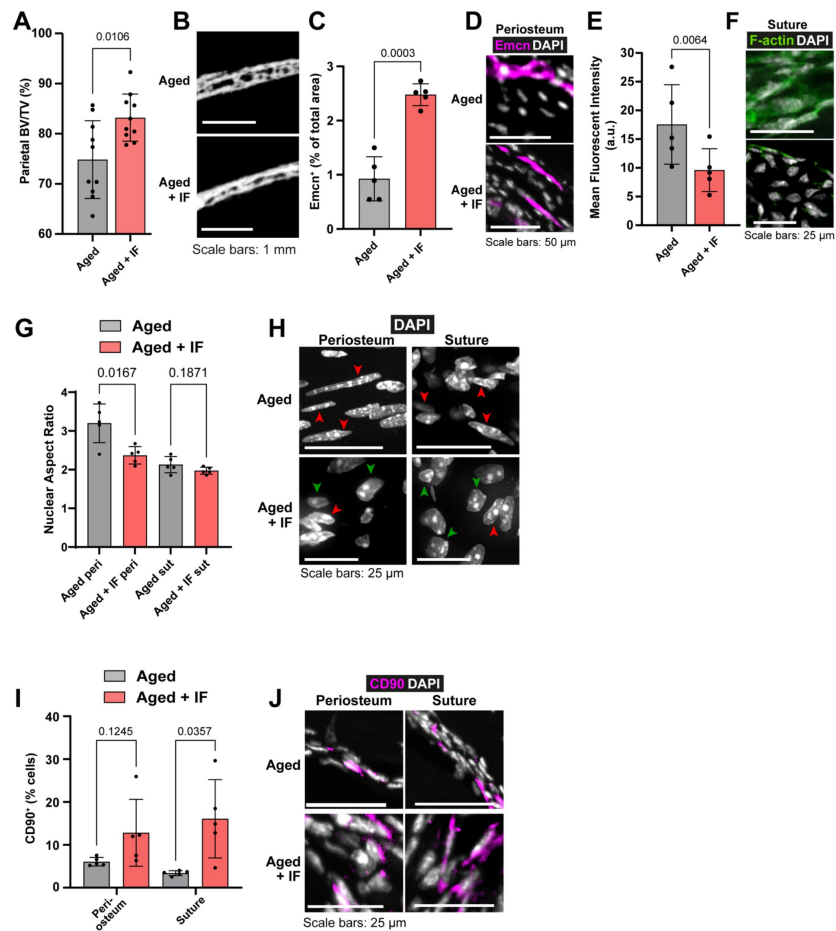


Figure 3

The impact of Intermittent Fasting on Aged bone structure and osteogenic tissues

(**A**) Parietal Bone Volume/Total Volume (BV/TV) and (**B**) representative transverse micro computed tomography (microCT) of Aged mice undergoing intermittent fasting. Statistical analysis: Welch's test in $n = 10$ mice per condition. (**C**) Mean cross sectional area taken up by Endomucin (Emcn)-positive vascular cells and (**D**) representative immunofluorescence staining of Emcn (magenta) and DAPI (grey) in the periosteum of Aged mice undergoing intermittent fasting. Statistical analysis: Welch's tests in $n = 5$ mice per condition. (**E**) Mean Phalloidin-488 fluorescence intensity and (**F**) representative images for filamentous actin (F-actin, green) and DAPI (grey) in the suture mesenchyme of Aged mice undergoing intermittent fasting. Statistical analysis: Welch's test in $n = 5$ mice per condition. (**G**) Nuclear aspect ratio (long axis divided by short axis) of cells within the periosteum and suture mesenchyme and representative DAPI-stained nuclei (**H**) of Aged mice undergoing intermittent fasting. Statistical analysis: Welch's tests in $n = 5$ mice per condition. Green and red arrows highlight rounded and elongated nuclei, respectively. (**I**) Percentage of CD90⁺ cells within the periosteum (left columns) and suture (right columns), expressed relative to the total cellular number (DAPI⁺) in Aged mice undergoing intermittent fasting. Representative images (**J**) show CD90 immunostaining (magenta) and DAPI (grey). Statistical analysis: Welch's tests in $n = 5$ mice. In panels **A**, **C**, **E**, **G** and **I**, columns in grey are data previously shown in [Fig. 1](#), reproduced here for reference. In panels **B**, **D**, **F**, **H**, **J**, Aged control representative images previously shown in [Fig. 1](#) are reproduced here for reference.

Simiczjjew et al., 2014 [\[1\]](#)). Zbtb16, encoding transcription factor PLZF, regulates osteogenesis by acting upstream of master osteoblast transcription factor Runx2 (**Supplementary Fig.9b** [\[1\]](#)) (Agrawal Singh et al., 2019; Liu and Lee, 2013 [\[2\]](#)).

In the suture mesenchyme, analysis of 297 significantly upregulated genes in Aged CD90⁺ cells during the IF diet compared to Aged AL control (**Fig. 4b** [\[1\]](#)) showed enrichment in 'Mitotic cell cycle' and 'Protein phosphorylation'. Notably, the proliferation marker Mki67 is significantly upregulated, with a threefold increase in Aged periosteum CD90⁺ cells during intermittent fasting compared to ad libitum diet.

In **Fig. 2a**, **a** [\[1\]](#) shift in the periosteal CD90⁺ was noted with aging. Aged AL had relatively fewer osteogenic Igf1^{high} cells, and more of Igf1^{low} cells. Contrary, Aged + IF CD90⁺ cells were split 70.3% Igf1^{high}, 29.7% Igf1^{low} (**Fig. 4c-d** [\[1\]](#)), positioning them between Young and Adult distributions. 2 of the 14 genes identified as conserved markers of periosteal CD90⁺ cell aging (**Supplementary Fig. 8b** [\[1\]](#)), Vcan, Mmp14, were significantly downregulated in Aged + IF. The proportion of 'proliferating' CD90⁺ suture cells increased in Aged + IF (**Fig. 4e-f** [\[1\]](#)). Further, 1 of the 32 conserved genes of the aging suture (**Supplementary Fig. 8c** [\[1\]](#)), actin-associated Tpm1 (Li et al., 2022 [\[3\]](#)), was significantly downregulated in Aged + IF. Wnt2, found to be downregulated with aging, is significantly upregulated in the suture CD90⁺ population of Aged + IF (source data_RNAseq table 10).

To validate diet-induced changes in CD90⁺ cells, we conducted immunostaining for protein markers in histological sections. ATPB, marker of ATP synthase, appeared elevated in periosteum and significantly increased in suture mesenchyme CD90⁺ cells following IF (**Fig. 4g-m** [\[1\]](#)). This suggests an increase in cellular ATP levels, which is supported by decreases in ATP-deficiency sensor phospho-AMPK in periosteal and suture CD90⁺ cells (**Fig. 4h**, **m** [\[1\]](#)). While proliferating cells were absent in Aged mice, we observed Ki67⁺ cells in samples of the periosteum of 3 out of 5 Aged + IF mice and in 4 out of 5 Aged + IF mice for suture mesenchyme (**Fig. 4i**, **m** [\[1\]](#)). Conversely, p16 was similar in periosteum and suture mesenchyme (**Fig. 4j**, **m** [\[1\]](#)). Wnt/βcatenin pathway activity, as measured by the relative nuclear localization of β-catenin, was significantly increased within the CD90⁺ cells of both the periosteum and suture (**Fig. 4k**, **m** [\[1\]](#)). Lastly, expression of the motility marker MMP2 in CD90⁺ cells was markedly increased over Aged controls in the periosteum, and just under statistical significance in the suture mesenchyme (**Fig. 4l**, **m** [\[1\]](#)).

Together, this data suggests that during IF, CD90⁺ cells, despite being Aged, are more capable of ECM modulation and migration, and are pro-osteogenesis.

Short term metabolic approaches rejuvenate the calvarial osteogenic compartments of Aged mice and promote bone repair

IF diet likely enhances energy supply to CD90⁺ cells and their function through various mechanisms. We next aimed to determine if specific underlying mechanisms alone could replicate the effects of IF on aged bone repair. Our analysis revealed that both aging and IF affect the NAD⁺ biosynthesis pathway and gut microbiota. Thus, we targeted these aspects to investigate their potential to enhance aged bone repair.

NMN supplementation improves bone repair in Aged mice

NAD⁺ is a multifunctional enzyme co-factor with roles including protein deacetylation, histone modification, intracellular signaling, and redox reactions (Covarrubias et al., 2021 [\[4\]](#)). It is also crucial in oxidative phosphorylation (Stein and Imai, 2012 [\[5\]](#)), cycling between oxidized and reduced forms (NAD⁺/NADH) in the mitochondria as part of the electron transport chain. Our analysis of inferred NAD⁺-dependent pathway activity (**Supplementary Fig.7e,f** [\[1\]](#)), live mitochondrial activity (**Fig. 1i-j** [\[1\]](#)), and markers of cellular energy levels (**Fig. 2f**, **g**, **l** [\[1\]](#))

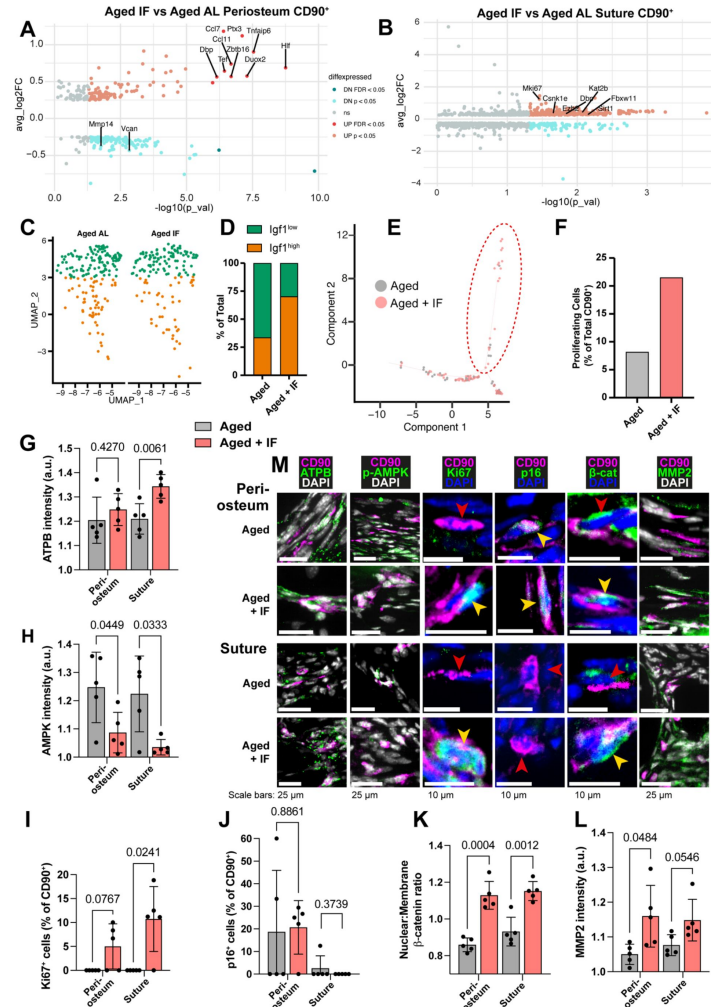


Figure 4

Transcriptomic and protein expression characterization of Aged CD90⁺ cells during intermittent fasting

(A) Volcano plot showing differentially-expressed genes in the comparison of the Aged CD90⁺ population in the periosteum and (B) suture during intermittent fasting (IF) vs the ad libitum (AL) diet. Statistical analysis: Wilcoxon Rank Sum test via the FindMarkers function in the Seurat R package. Upregulated (pink: *p* < 0.05; red: FDR-adjusted *p* < 0.05), downregulated (cyan: *p* < 0.05; teal: FDR-adjusted *p* < 0.05), and nonsignificant (ns) genes are displayed on axes of log fold expression (IF over AL) (y, avg_log2FC) against *p* transformed as its negative log value. Periosteum: Aged IF (n = 194 cells); Aged AL (n = 242 cells). Suture: Aged IF (n = 79 cells); Aged AL (n = 61 cells). (C) UMAP-plot showing the distribution of a subpopulation of CD90⁺ periosteal cells during intermittent fasting in Adult and Aged mice. Ad libitum (AL) cells are data previously shown, reproduced here for reference. (D) Percentage of the total cells categorized as either *Igf1*^{high} or *Igf1*^{low}. Aged + IF: *Igf1*^{high} (n = 104), *Igf1*^{low} (n = 44). (E) Plot (Monocle package) of suture CD90⁺ populations. (F) Percentage of CD90⁺ categorized as 'proliferating' (inside red dashed ellipse) vs 'non-proliferating' (outside ellipse) in Aged mice undergoing intermittent fasting. Ad libitum (AL) cells and chart are data previously shown, reproduced here for reference. Statistical analysis: two-tailed binomial test based on expected observed proportions between groups. Adult + IF: proliferating (n = 30 cells), non-proliferating (n = 86); Aged + IF: proliferating (n = 17), non-proliferating (n = 62). (G) Normalized intensity staining for ATPB in CD90⁺ cells. (H) Normalized intensity staining for phospho-AMPK in CD90⁺ cells. (I) Proportion of CD90⁺ cells that are Ki67⁺. (J) Proportion of CD90⁺ cells that are p16⁺. (K) Ratio of Nuclear:Membrane β -catenin staining in CD90⁺ cells. (L) Normalized intensity staining for MMP2 in CD90⁺ cells. For all plots (G-L), Aged + IF (red), Aged control (grey), periosteum (2 left columns), suture mesenchyme (2 right columns), statistical testing: Welch's tests in n = 5 mice per condition. (M) Representative images depicting CD90 (magenta), DAPI (grey/blue as indicated by column), and functional markers (green) as indicated by column). Yellow arrows indicate marker-high/positive cells; red arrows indicate marker-low/negative cells. In panels C-M, data and representative images of Aged controls from previous figures are reproduced here for reference.

demonstrated distinct changes to metabolism with age. Yet, little is known about the effect of NAD⁺-coupled cellular metabolism on the composition of aging calvarial periosteum and suture mesenchyme and its influence over Aged bone repair.

To increase cellular levels of NAD⁺ systemically, we orally dosed Aged mice with nicotinamide mononucleotide (NMN), an immediate precursor to NAD⁺ in the ‘salvage’ pathway (Yang and Sauve, 2016). During a pilot study involving oral NMN supplementation (600 mg/kg, 3 times within 1 week) in Adult mice, the mitochondrial membrane potential, as measured by TMRM staining, within the cells of the periosteum was significantly increased (Fig. 5a, b). NMN was then administered to Aged mice for the 2 weeks prior and 2 weeks following calvarial defect surgery.

Parietal bone BV/TV was significantly increased in Aged animals that received NMN (Aged + NMN) (Fig. 5c–d). Remarkably, Aged bone repair was significantly increased to 54.32% with NMN with inactive bandages (Fig. 1o–p), marking a three-fold improvement over control Aged mice. Adding Wnt3a-bandage increased Aged + NMN repair to 71.18%, resulting in over 2.71 times greater area of new bone compared to Aged controls + Wnt3a-bandage, akin to repair in Adult mice + Wnt3a-bandage.

Next, we probed how NMN affects CD90⁺ cells in the osteogenic compartments. In the suture mesenchyme, NMN treatment significantly increased CD90⁺ cells compared to Aged control (Fig. 5e–f). In the periosteum, CD90⁺ populations were unchanged. Additionally, compared to Aged controls, NMN supplementation significantly increased the vascularity of the periosteum (Fig. 5g–h), and approximately halved actin cytoskeletal staining intensity in the suture mesenchyme (Fig. 5i–j). Nuclei in the suture and periosteum were rounder and less elongated than Aged controls (Fig. 5k–m), making them morphologically resemble Adult nuclei.

Taken together, NMN treatment increased periosteal vascularity, impacted aspects of the mechanobiology of the periosteum and suture mesenchyme, leading to a significant improvement in bone repair in Aged mice.

Cyclical nutrient deprivation and NMN enhance mitochondrial function in osteogenic cells in vitro

Next, we aimed to model aspects of the effects of intermittent fasting and NMN supplementation on osteogenic cells, employing a cyclical nutrient deprivation regimen or NMN supplementation *in vitro* to elucidate aspects of their mechanisms of action. While this approach allows us to study specific cellular responses, it is crucial to recognize that it cannot fully replicate the complex systemic effects of *in vivo* intermittent fasting.

We investigated the impact of these interventions on mitochondrial function in the MC3T3 calvarial osteoblastic cell line, a model commonly utilized to study osteogenesis *in vitro* and known for its ability to contribute to bone formation *in vivo* (Horowitz et al., 1994; Paine et al., 2018; Wang et al., 1999).

Alternating serum deprivation every hour (cyclical nutrient deprivation) in glucose-free medium induced the expression of genes related to mitochondrial biogenesis, fusion, mitophagy, and the unfolded protein response (Fig. 6a). This was accompanied by increased mitochondrial filament length and membrane potential, indicating enhanced mitochondrial function (McCarron et al., 2013; Westermann, 2012) (Fig. 6b–e). These findings are consistent with *in vivo* observations of improved mitochondrial function in animals subjected to intermittent fasting. While cyclical nutrient deprivation enhanced mitochondrial membrane potential, it also reduced

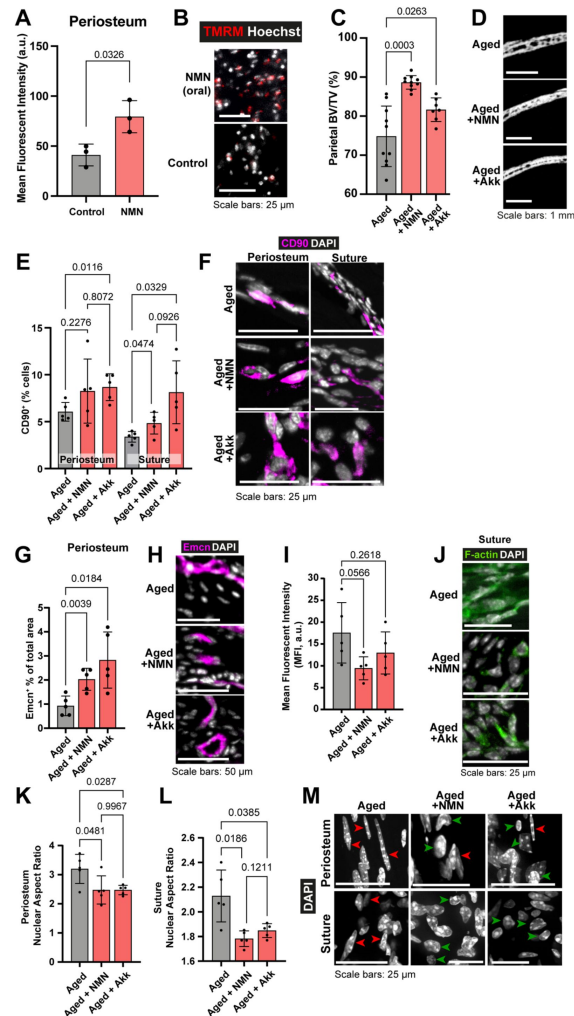


Figure 5

The impact of Nicotinamide mononucleotide (NMN) and Akkermansia muciniphila (Akk) supplementation on Aged bone structure and osteogenic tissues

(A) Mean intensity and (B) representative images of live TMRM staining (red) and Hoechst (grey) in Adult mice controls and Adult mice dosed with 600 mg/kg NMN by oral gavage, samples of periosteum *ex vivo*. Statistical analysis: Welch's test in tissue from $n = 3$ mice per condition. (C) Parietal Bone Volume/Total Volume (BV/TV) and (D) representative transverse micro-computed tomography (microCT) sections in Aged mice supplemented with NMN or Akkermansia. Statistical analysis: Welch's tests in $n = 10$ (Aged, Aged + NMN) and $n = 7$ (Aged + Akk) mice. (E) Percentage of CD90⁺ cells within the periosteum (left columns) and suture (right columns), expressed relative to the total cellular number (DAPI⁺) in Aged mice supplemented with NMN or Akkermansia. Representative images (F) show CD90 immunostaining (magenta) and DAPI (grey). Statistical analysis: Welch's tests in $n = 5$ mice per condition. (G) Mean cross sectional area taken up by Endomucin (Emcn)-positive vascular cells and (H) representative immunofluorescence staining of Emcn (magenta) and DAPI (grey) in the periosteum of Aged mice supplemented with NMN or Akkermansia. Statistical analysis: Welch's tests in $n = 5$ mice per condition. (I) Mean Phalloidin-488 fluorescence intensity and (J) representative images for filamentous actin (F-actin, green) and DAPI (grey) in the suture mesenchyme of Aged mice supplemented with NMN or Akkermansia. Statistical analysis: Welch's tests in $n = 5$ mice per condition. (K-M) nuclear aspect ratio (long axis divided by short axis) of cells within the (K) periosteum and (L) suture in Aged mice supplemented with NMN or Akkermansia, and (M) Representative DAPI-stained nuclei. Statistical analysis: Welch's tests in $n = 5$ mice per condition. Green and red arrows highlight rounded and elongated nuclei, respectively. (N) The relative abundance of bacteria identified as genus Akkermansia in 16S metagenomic profiling of Aged mice undergoing intermittent fasting vs. Aged controls. Statistical analysis: Welch's test in (Aged, $n = 3$; Aged + IF, $n = 4$) mice per condition. In panels C- M, data and representative images of Aged controls from previous figures are reproduced here for reference.

overall cellular ATP levels. However, inhibiting mitochondrial ATP production with oligomycin (a Complex V inhibitor)(Devenish et al., 2000 [↗](#)) further diminished ATP levels, suggesting that mitochondrial respiration is a primary source of ATP in these cells (**Fig. 6f** [↗](#)).

Recognizing the role of mitochondrial dysfunction in aging, we investigated the impact of NMN on MC3T3 cells challenged with oligomycin and cultured in glucose and pyruvate-free medium. NMN treatment mitigated oligomycin-induced ATP loss and ROS production, enhanced cell proliferation, and prevented the loss of mitochondrial membrane potential (**Fig. 7a-c** [↗](#)). Holotomography microscopy revealed that NMN preserved the "rod-like/filament" morphology of mitochondria in oligomycin-treated cells (**Fig. 7d-e** [↗](#)). Furthermore, NMN restored the expression of genes related to mitochondrial biogenesis and dynamics, the unfolded protein response, and mitophagy, which were adversely affected by oligomycin (**Fig. 7f** [↗](#)). Ultimately, NMN enhanced the growth and differentiation potential of oligomycin-treated cells, as evidenced by increased ALP activity and mineralization (**Fig. 7g** [↗](#)).

In summary, our findings suggest that both cyclical nutrient deprivation and NMN supplementation can enhance mitochondrial function in osteogenic cells. NMN appears to protect against mitochondrial dysfunction induced by Complex V inhibition, highlighting its potential in mitigating the decline in osteogenic cell function.

Akkermansia muciniphila supplementation improves bone repair in Aged mice

The 'gut-bone axis' links food intake, gut microbiome and bone metabolism via circulating metabolites(Villa et al., 2017 [↗](#)). Bacteria in the genus *Akkermansia* appeared to be elevated during Aged + IF compared to the AL control diet (**Fig. 5n** [↗](#)), consistent with previous studies that found similar results in young mice and human(Li et al., 2020 [↗](#); Özkul et al., 2019 [↗](#)).

Akkermansia muciniphila is an anaerobic gut bacterium responsible for mucin turnover(Derrien et al., 2008 [↗](#)), and a regulator of the intestinal mucus environment(Reunanen et al., 2015 [↗](#)). The *Akkermansia* population exerts wider effects on the gut microbiome, and has been studied for its roles in metabolic health(Dao et al., 2016 [↗](#); Leite et al., 2021 [↗](#)). We cultured *Akkermansia muciniphila* (Akk) under anaerobic conditions and calculated the viability of cryopreserved aliquots. Doses of live Akk (2×10^8 C.F.U.) were given by oral gavage three times per week in Aged mice for the 2 weeks prior and 6 weeks following calvarial defect surgery.

Our metagenomic analysis reveals that Firmicutes:Bacteroides ratio, indicator of microbiome composition(Magne et al., 2020 [↗](#)), is significantly increased in Aged mice with *Akkermansia* supplementation (Aged + Akk) (**Supplementary Fig. 10a** [↗](#)). Furthermore, the proportion of gut bacteria belonging to family *Lachnospiraceae*, indicative of the capacity of the gut bacteria to produce short chain fatty acids (SCFAs), is significantly elevated in Aged + Akk (**Supplementary Fig. 10b** [↗](#)). SCFAs enter the host bloodstream and exert epigenetic effects primarily through histone modification, which might have implication on cell function and differentiation(Licciardi et al., 2011 [↗](#)).

Similar to NMN treatment, parietal bone BV/TV was significantly increased Aged + Akk (**Fig. 5c-d** [↗](#)). Aged bone repair was significantly increased to 43.76% with Akk and inactive bandages (**Fig. 10-p** [↗](#)). In combination with the Wnt3a bandage effect, bone repair in Aged + Akk further increased to 64.84%. This level of bone repair is similar to that of Young mice with Wnt3a-bandage (**Supplementary Table 1** [↗](#)). Aged + Akk also produced striated bone, similar in structure to Young mice, under both Wnt3a and control bandages (**Supplementary Fig. 3d** [↗](#)).

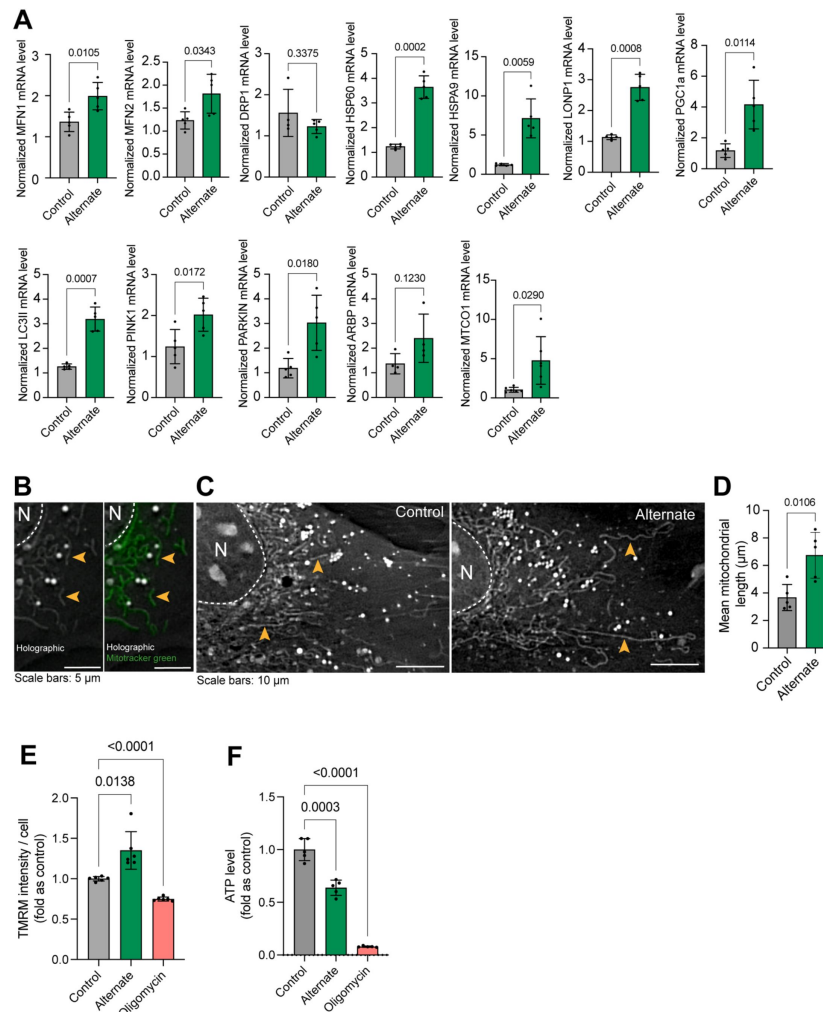


Figure 6

Effect of hour-based alternate serum deprivation on mitochondria in MC3T3 cells in glucose-free media.

(A) Effect of alternate serum deprivation on gene expression implied in mitochondrial dynamics (MFN1, MFN2, DRP1), unfolded protein response (mtUPR) (HSP60, HSPA9, LonP1), biogenesis (PGC1a), structure (ARBP), mitophagy (LC3II PINK1, PARKIN), and mitochondrial respiratory complex IV (MTCO1). Statistical analysis: Welch's test, $n \geq 4$ /group. **(B)** Representative holographic images of mitochondrial shape (orange arrowheads), confirmed by Mitotracker green staining. **(C-D)** Effect of alternate serum deprivation on mitochondrial length. Statistical analysis: Welch's t-test of $n = 5$ /group, total mitochondria = 425. **(E)** Effect of alternate serum deprivation and oligomycin on mitochondrial membrane potential. Statistical analysis: Welch's t-test of $n = 6$ /group with at least 481 cells/group. **(F)** Effect of alternate serum deprivation and oligomycin on ATP level. Statistical analysis: Welch's t-test of $n \geq 5$ /group.

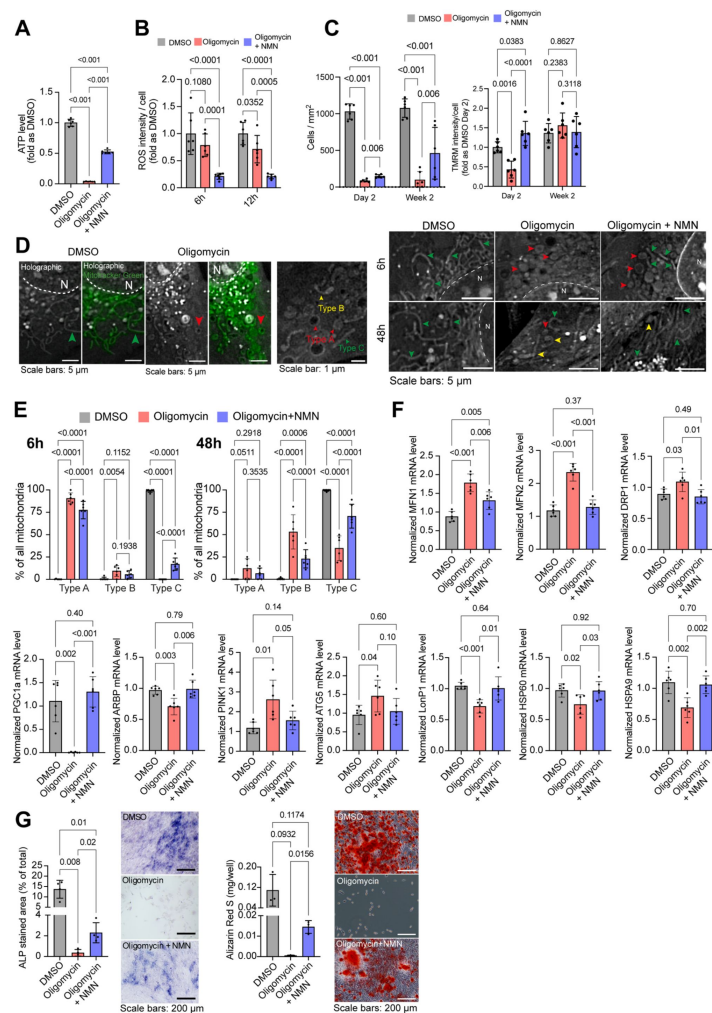


Figure 7

Effect of NMN on oligomycin-induced mitochondrial damage in glucose-free MC3T3 cells.

(A) Effect of oligomycin ± NMN on ATP level in MC3T3 cells. Welch's test, $n=6$ /group. (B) Effect of oligomycin ± NMN on ROS level MC3T3 cells. Welch's test, $n=6$ /group. (C) Effect of oligomycin ± NMN on cell density (left) and mitochondrial membrane potential (right) over time in MC3T3. Welch's test, $n=6$ /group. (D) Representative images of mitochondrial shape alteration by oligomycin exposure ± NMN supplementation. Dashed white line = nuclear membrane, N = nucleus. Observation by holotomography (grey levels) and confirmed with Mitotracker Green (green). Red arrowheads = Type A ("round"), yellow arrowheads = Type B ("intermediate"), and green arrowheads = Type C ("rod-like/filamentous") mitochondria. (E) Quantification of the effect of oligomycin ± NMN on mitochondrial shape. Each data point is a cell with at least 30 mitochondria. Welch's test $n = 6$ cells / group, total mitochondria=1821. (F) Effect of oligomycin ± NMN on gene expression implied in mitochondrial dynamics (MFN1, MFN2, DRP1), biogenesis (PGC1a), structure (ARBP), mitophagy (PINK1, ATG5), and unfolded protein response (mtUPR) (LonP1, HSP60, HSPA9). Welch's test, $n \geq 5$ /group. (G) Effect of oligomycin ± NMN on osteoblastic differentiation. Staining of alkaline phosphatase (ALP) activity and mineralization (Alizarin Red S; ARS) of the matrix after two weeks of differentiation. Welch's test, $n=4$ (ALP) or 3 (ARS) / group.

Aged + Akk led to a significant increase in CD90⁺ cells in the periosteum and suture mesenchyme (Fig. 5e-f) and a threefold increase in the Emcn⁺ vascularity area of the periosteum (Fig. 5g-h). Compared to Aged controls, in Aged + Akk, actin cytoskeletal staining intensity appeared reduced (not significant) in the suture mesenchyme (Fig. 5i-j), and nuclei in both the suture and periosteum appeared rounder and less elongated (Fig. 5k-m). Accordingly, the nuclear morphology of Aged + Akk cells resembles that of younger mice.

Elevated osteoclast activity associated with enhanced bone health in Aged mice following interventions

To investigate the impact of age and interventions on bone remodeling, we assessed osteoclast activity across different age groups (Young, Adult, and Aged) and compared aged animals who underwent either IF or Akk or NMN interventions to untreated controls. The tartrate-resistant acid phosphatase (TRAP) assay (Yang et al., 2019) results revealed an expected increase in osteoclast activity with age (Supplementary Fig. 10c-d). Interestingly, Aged mice treated with IF, NMN, or Akk exhibited significantly higher TRAP activity compared to the untreated aged control group. This finding suggests that these interventions not only enhance bone formation but also promote the removal of old or damaged bone by stimulating osteoclast activity. This coordinated increase in both bone formation and resorption likely contributes to a more youthful and efficient remodeling process, ultimately leading to improved bone health and repair in aged mice.

In summary, we have revealed a multifaceted decline in CD90⁺ cells with aging, which can be locally counteracted by a Wnt3a-bandage or systemically enhanced through an IF diet. Moreover, systemic interventions, acting via the gut and on shorter terms, can improve CD90⁺ populations, osteogenic compartments, and significantly enhance bone repair in Aged mice. Combining the Wnt3a-bandage with systemic interventions targeting cellular energy supply, proliferation, migration, and Wnt signaling presents coordinated and clinically relevant approaches, promising rejuvenation of Aged bone repair comparable to younger individuals.

Discussion

Here we report on systemic interventions that can reverse aspects of the aging osteogenic tissue to promote effective calvarial bone repair (Supplementary Fig. 11).

While extensive research focuses on long bones (Yates et al., 2007), there is a significant gap in understanding aging in flat bones. Our study on the calvaria demonstrates a clear decline in both bone structure and healing capacity with age. Focusing on osteogenic tissues, we observed a progressive decline in periosteal blood vessel density beginning in adulthood. Microvascular rarefaction is increasingly recognized as a driver, rather than merely a consequence, of hallmarks of cellular aging, such as cellular senescence (Goligorsky, 2010; Lähteenvuo and Rosenzweig, 2012). Indeed, previous studies inducing vasculature loss in young animals accelerates cellular aging (Chen et al., 2021), while mitigating this vascular decline preserves tissue health and delays aging (Grunewald et al., 2021).

Aging bone tissues are inherently more hostile and pro-apoptotic (Almeida, 2012), leading to limited success with cell-based repair approaches in aged individuals (Hirata et al., 2022; Liu et al., 2018). Stiffness increases with aging in various tissues (Ferrari and Pesce, 2021; Lim and Ito, 2022; Segel et al., 2019), impacting cytoskeletal configuration (Khan et al., 2020; Pongkitwitoon et al., 2016) and significantly affecting osteogenic differentiation and senescence *in vitro* (Liu et al., 2022). Our study shows increased actin expression and elongated nuclear shape in aging osteogenic compartments, alongside elevated levels of Mmp9 and Mmp14, linked to stiffer osteogenic tissues (Tang et al., 2013; Zhu et al., 2020). These changes are associated with a decline in both the number and function of osteoprogenitors, including impaired proliferation,

mitochondrial energy metabolism, Wnt/ β -catenin pathway activation, migration, and ultimately, bone defect repair. While Wnt-bandage treatment increases the number of osteoprogenitors in the defect site across all age cohorts, it only modestly enhances bone repair in aged animals compared to the healing observed in adult mice.

We show that interventions improving whole-body energy metabolism can enhance aged bone health and repair. While IF is effective in delaying aging in adult flies, animals need to switch back to ad libitum feeding at older ages (Catterson et al., 2018 [DOI](#)), and increased protein intake is suggested for optimizing health span and longevity in aged mice and humans (Levine et al., 2014 [DOI](#)). Our findings, however, reveal a distinct paradigm for IF in aged bone repair. Specifically, short-term IF initiated post-injury robustly enhances bone healing in aged animals, comparable to younger counterparts.

Mechanistically, IF exerts its effects through several pathways, including enhanced stress resistance, increased reliance on lipid fuel utilization, and activation of proteostatic mechanisms, all contributing to delayed aging (Longo and Anderson, 2022 [DOI](#)). Although we did not observe distinct changes in overall signature of lipid metabolism or glycolysis in osteogenic tissues during aging or IF, we did find a decline in mitochondrial function and ATP supply with age. Our *in vitro* studies showed that alternating cycles of serum deprivation and supply in osteogenic cells induced genes related to the unfolded mitochondrial protein response, mitophagy and mitochondrial fission associated with increased mitochondrial membrane potential and mitochondrial fission, indicating improved function (McCarron et al., 2013 [DOI](#); Westermann, 2012 [DOI](#)).

Our transcriptomic analysis shows that *Mmp14* is reduced during IF in the periosteum. In the suture mesenchyme, *Piezo1*, a gene known to mediate mechanically induced aging (Ren et al., 2022 [DOI](#); Segel et al., 2019 [DOI](#)), was also significantly reduced in aged mice undergoing IF. Along with the reversal of nuclear cell morphology and levels of F-actin, these results indicate that IF also impacts the mechanobiology of the tissue, making it resemble that of younger animals.

IF also led to beneficial shifts in gut microbiota composition, including elevated levels of *Akkermansia muciniphila* and increased expression of genes involved in NAD⁺ biosynthesis. Subsequently, we show that short-term supplementation with *Akkermansia muciniphila* or NMN in aged mice significantly improve periosteal blood vessel density, osteoprogenitor number and function, osteoclast activity and enhance bone health and repair.

Although IF, Akk or NMN can act through many pathways and cellular processes, previous studies highlighted their beneficial effect on metabolism particularly their influence on NAD-dependent pathways. Mechanistically, our *in vitro* studies using calvarial osteogenic cells revealed that NMN protects against mitochondrial damage, often associated with aging, promotes healthy mitochondrial morphology and function, and enhances osteogenesis.

The effects of the three metabolic approaches on bone health and repair have not been studied in the context of already-aged individuals. NMN shows protective effects against bone health decline, but in long-term studies that begin dosing well before old age (Kim et al., 2021 [DOI](#); Mills et al., 2016 [DOI](#); Song et al., 2019 [DOI](#)). The impact of *Akkermansia muciniphila* on bone health or healing has only been investigated in young adult models with subcritical defects (Lawenius et al., 2020 [DOI](#); Liu et al., 2020 [DOI](#)). IF diet has been investigated for skin wounding (Luo et al., 2020 [DOI](#)) in young animals but not for bone repair. We present key progress in developing three clinically relevant approaches for bone tissue regeneration in elderly patients. First, we describe the effects of NMN or *Akkermansia* dosing, or IF diet on bone tissues in aged animals that have already experienced severe skeletal decline. Second, our rejuvenation protocols are applied for shorter terms, and even post-injury with IF diet.

This study offers valuable insights into bone aging and rejuvenation strategies. However, future research should address several limitations. Further investigation is needed on the mechanisms of action of IF, NMN, and Akkermansia, optimal intervention timing and duration, and synergistic effects of combined approaches. While this study focused on a common osteoprogenitor, investigating the roles of specific skeletal stem cell populations and other cells within the osteogenic compartments (including immune cells) in flat bones represents a compelling frontier.

Our findings suggest broader implications beyond bone health, with these rejuvenation strategies potentially translating to other aging organs and injuries. Rigorous investigation of dosage, safety, and long-term effects will be essential for human applications.

Methods

Wnt3a bandage manufacture

Polycaprolactone (PCL)-based polymer film (Poly-Med) was plasma-treated using a plasma etching electrode in oxygen gas at a flow of $20 \text{ cm}^3 \text{ min}^{-1}$ with a radio frequency plasma generator (frequency 13.56 MHz, power 50 W, Diener electronic Zepto-W6) at 0.2 mbar pressure for 3 min. Treated films were immediately incubated in 5% (3-Aminopropyl)triethoxysilane (APTES, Sigma, 440140) in ethanol (dark, room temperature, 2 hours), before washing twice with 100% ethanol (room temperature, 10 minutes), and allowed to air dry. Circular bandages were cut using a 5mm biopsy punch (Stiefel) and placed in 10 mg/ml succinic anhydride (Sigma Aldrich, 239690-50G) in 0.4M borate buffer (Sigma Aldrich, B7660) pH=9 (room temperature, 30 minutes). Bandages were twice washed in 25mM Morpholineethanesulfonic acid (MES) buffer pH 5 (Sigma Aldrich, M3671), then incubated in a MES buffer solution containing 50 mg/ml N(3Dimethylaminopropyl)-N'-ethylcarbodiimide hydrochloride (EDC, Sigma Aldrich, E7750) and 50 mg/ml NHydroxysuccinimide (NHS, Sigma Aldrich, 56480) (room temperature, 30 minutes), and washed twice again in MES buffer. Bandages are individually placed on a sterile flat surface and briefly allowed to air-dry. 300ng Wnt3a (R&D, 1324-WN-010) was diluted to 23 μL in MES buffer, and this liquid added as a droplet on top of the bandage for 1 hour at room temperature.

For inert control bandages (inactive-bandages), the bandages were then placed into a 96 well plate and washed with PBS. 20 mM Dithiothreitol (DTT, Roche, 10197777001) in 50 μL ddH₂O was then added and incubated (37°C, 30 minutes). Active Wnt3a-bandages were also placed in the 96 well plate, and then all bandages were washed with PBS, and quenched with 2% Fetal Bovine Serum (FBS, Sigma Aldrich, F7524) in Dulbecco's Modified Eagle Medium (DMEM, Thermo Fisher, 41966) (room temperature, 15 minutes) in preparation for their surgical implantation.

Wnt3a-bandage testing

The Wnt/ β -catenin pathway activation of Wnt3a-bandages was confirmed as previously published (Lowndes et al. 2017 [\[1\]](#)). Wnt3a-bandages were tested for Wnt/ β -catenin pathway activation using Comma-D Beta cells harboring GFP expression under the control of a TCF-LEF promoter (Lowndes et al. 2016 [\[2\]](#)) (Supplementary Fig. 3a, [c](#) [\[3\]](#)). Similarly, LS/L cells harboring luciferase under control of a 7xTCF promoter were used to quantitatively test Wnt3a-bandages, as described previously (Okuchi et al. 2021 [\[4\]](#)) (Supplementary Fig. 3b [\[5\]](#)).

Animal husbandry/housing

Female C57BL6/J mice (Charles River, strain 632) were housed in conditions that comply with the Animals (Scientific) Procedures Act 1986 (ASPA). Mice were group housed (2-5 mice/cage) under 7am-7pm lighting and fed with standard mouse food pellets (Lab diet Picolab Rodent Diet 20 EXT Irradiated). During intermittent fasting, food was removed from cages at 10am on Monday,

Wednesday, and Friday, and returned at 10am on Tuesday, Thursday, and Saturday, such that the maximum time without food was 24 hours. Intermittent fasting started the day after the surgery and was performed for 10 weeks.

Calvarial Defect

All steps of surgery were carried out under aseptic conditions. Mice were intraperitoneally anaesthetized with 10 μ L/g bodyweight of anesthesia cocktail (7.5 mg/ml Ketamine hydrochloride (Ketavet) and 0.1 mg/ml Medetomidine hydrochloride (Domitor) in 0.9% sterile saline). Loss of consciousness was confirmed with lack of pedal and tail reflexes. The top of the head was shaved, cleaned, and prepared for surgery, before a ~8 mm central incision was made in the skin with surgical scissors. The periosteum was gently pushed towards the outer edges of the calvaria, and the location of the sutures noted. A 2 mm diameter full thickness defect was drilled into the middle of the parietal bone on both sides of the skull, ensuring irrigation was applied to limit heat buildup. A small amount (~10 μ L) of 1 mg/ml collagen gel (Corning, 354249) (diluted with DMEM and set with 3 mM sodium hydroxide (Sigma Aldrich, S5881)) was placed into the defect to exclude trapped air, and the required bandages were placed face down over the defect. Small dots of glue (3M, Vetbond) were applied at the edges of the bandages to affix them to the surface of the skull. The skin was then approximated and glued to ensure closure of the wound. Mice were administered 10 μ L/g bodyweight of the recovery cocktail intraperitoneally (0.125 mg/ml Atipamezole hydrochloride (Antisedan) and 0.01 mg/ml Buprenorphine hydrochloride (Vetergesic) in 0.9% sterile saline) and placed in a warmed cage (35 °C). Mice typically became conscious within 20 minutes, began moving around the cage within 40 minutes, and were feeding within 2 hours. Mice showed no signs of distress or pain the following day and were externally monitored daily for the healing of the skin wound. Following surgery, mice were maintained under standard conditions for 10 weeks, and group housed (2-5 mice/cage) where possible.

NMN supplementation

600 mg/kg bodyweight doses of β -nicotinamide mononucleotide (NMN, APExBIO, B7878), dissolved in water were given by flexible-tipped oral gavage 3 times per week, not on consecutive days, from 2 weeks before surgery to 2 weeks after, a total of 4 weeks.

Anaerobic Culture of *Akkermansia muciniphila*

Preserved *Akkermansia muciniphila* (Collection de l'Institut Pasteur, CIP107961T) was resuspended and cultured in Gifu's Anaerobic Medium (GAM) broth (Nissui Pharmaceutical, 05422) in an anaerobic environment (80% Nitrogen, 10% Carbon dioxide, 10% Hydrogen). Aliquots were cryopreserved in 50% glycerol (Sigma Aldrich, G2025) in PBS. Post-thawing viability was determined by serial dilution of cryopreserved aliquots, plating onto GAM-Agar (Sigma Aldrich, 01916) plates, and 24-hour anaerobic culture.

Akkermansia muciniphila supplementation

Cryopreserved doses of 2.0×10^8 C.F.U. *Akkermansia muciniphila* were warmed to ~ 30°C and administered to Aged mice by flexible-tipped oral gavage 3 times per week, not on consecutive days, from 2 weeks before surgery to 6 weeks after, a total of 8 weeks.

Microbiome Profiling

Feces samples were freshly collected and snap-frozen in liquid nitrogen. Samples were submitted for full length (V1-V9) 16S metagenomic profiling. Amplification with KAPA HiFi HotStart ReadyMix PCR Kit (KAPA) and 16S barcode primers (PacBio) preceded library preparation SMRTbell Express Template Prep 2.0 (PacBio). Samples were sequenced via the Illumina MiSeq to generate amplicon sequence data specifying the V3-V4 regions of the 16S rRNA ribosomal subunit. The metagenomic analysis was conducted using the amplicon sequence variant (ASV) method of

taxonomic classification facilitated by DADA2 (Callahan et al. 2016 [DOI](#)) incorporated by the Qiime2 package (Bolyen et al. 2019 [DOI](#)). Quality control metrics of the amplicon data was visualized by FastQC (Andrews S. 2015 [DOI](#)). The data was cleaned, via removal of adapters and low quality sequences, using Trimmomatic (Bolger et al. (2014) [DOI](#)) and Qiime2.

Ethical Approval for animal experimentation

All procedures involving animals were carried out by holders of Personal Licences (PIL) under a Project Licence (PPL, P8F273FDD, S.J.H.) shaped by ARRIVE and NC3Rs guidelines. The PPL was approved by the Animal Welfare and Ethical Review Body (AWERB) at King's College London and granted by the Home Office under the Animals (Scientific) Procedures Act 1986 (ASPA) and Amendment Regulations 2012.

Micro-Computed Tomography

Mice were culled by cervical dislocation and their tissues immediately collected. Whole calvarial samples were fixed in 4% PFA (room temperature, overnight) and washed in PBS. Fixed samples were scanned on a μ CT50 micro-CT scanner (Scanco) in 19 mm scanning tubes, using voxels of side length 10 μ m, X-ray settings of 70 kVp and 114 μ A and using a 0.5 mm aluminum filter to attenuate harder X-rays. Scans were reconstructed and calibrated using known densities of hydroxyapatite (HA) between 0 and 790 mg HA cm⁻³, with absorption values expressed in Hounsfield Units (HU). 3D volumes were constructed from raw data in SCANCO Visualizer 1.1, and signal of less than 250 mg HA cm⁻³ was considered non-mineralized and was set as the black level in displayed images.

Sample Processing and Cryosectioning

Fixed samples were decalcified in 10% formic acid (room temperature, overnight), then washed in PBS. Dehydration was carried out in 30% sucrose/PBS (room temperature, overnight), followed by 30% sucrose/30% OCT/PBS (room temperature, overnight). Samples were embedded in 100% OCT solidified with dry ice and stored at -80°C. Cryosections were cut at 14 μ m thickness using a cryostat microtome at 23°C. Cut sections were stored at -20°C.

Fluorescence Immunostaining

Sections were rehydrated to ddH₂O. Antigen retrieval was performed with 0.05% v/v Tween-20 (Sigma Aldrich, P9416) in 10 mM Tris (Sigma Aldrich, 93362), 1 mM EDTA (Sigma Aldrich, E6758), pH=9.0 (40°C, 30 minutes). Sections were then permeabilized with 0.1% Triton-X (Sigma Aldrich, X100) in 1% Bovine Serum Albumin (BSA, Sigma Aldrich, A2153) in phosphate buffered saline (PBS, Sigma Aldrich, P4417) (1% BSA/PBS) (room temperature, 5 minutes), washed with 1% BSA/PBS, and individual staining regions were designated with hydrophobic delimiting pen (Agilent, S200230-2).

Primary antibodies were diluted in 0.05% Triton-X in 1% BSA/PBS and incubated in a humid chamber (4°C, overnight). Triplicate washes of 0.1% Triton-X in 1% BSA/PBS were carried out (room temperature, 10 minutes). Secondary antibodies were diluted in 0.05% Triton-X in 1% BSA/PBS, with 5 μ g/ml DAPI (Sigma Aldrich, D9542), and 1:400 dilution Phalloidin-488 (Thermo Fisher, A12379) and incubated in a humid chamber (room temperature, 1 hour). Triplicate washes of 0.1% Triton-X in 1% BSA/PBS were carried out (room temperature, 10 minutes) before mounting with aqueous mounting media (Abcam, ab128982).

Immunostained sections were imaged with an inverted spinning disk confocal microscope (Nikon, NISElements Viewer 5.2 software). Optimal z-axis step size was determined according to the Nyquist criteria for sampling. Images were post-processed and analyzed through ImageJ/FIJI.

Primary antibodies

Anti-Endomucin (Emcn) 1:100 (Santa Cruz, sc-65495) Anti-CD90/Thy1, 1:100 (Abcam, ab3105) Anti-phosphoAMPK (Thr183, Thr172), 1:100 (Thermo Fisher, 44-1150G) Anti-ATPB, 1:100 (Abcam, ab128743) Anti-Ki67, 1:100 (Thermo Fisher, MA5-14520) Anti-p16INK4a, 1:100 (Abcam, ab211542) Anti-beta-catenin 1:100 (Abcam, ab32572) Anti-MMP2, 1:100 (Abcam ab92536) Anti-CD31, 1:100 (Novus Biologicals, NB100-2284)

Secondary antibodies

Chicken anti-Rat-647 1:1000 (Thermo Fisher, A21472) Donkey anti-Rabbit-555 1:1000 (Thermo Fisher, A32816)

Fluorescence thresholding quantification

See *Supplementary Figure 1a*. Sample tissue sections were immunostained alongside controls that lacked primary antibody: the ‘no primary control’. These control samples were imaged using the same microscope settings as the experimental samples, and maximum projections were calculated to reduce volumetric images to 2D. The maximum pixel intensity in the control images was used to threshold the experimental samples, thereby excluding signal arising from non-specific staining by fluorescent secondary antibodies. Cellular localization of the remaining signal allowed designation of cells positive and negative for the antibody targets of interest.

For quantitative measurements of F-actin/Phalloidin-488 signal intensity (e.g. **Figure 1h** [↗](#)), sum-projections were taken to reduce volumetric 3D images to 2D space. In these quantifications, mean sum pixel intensity was measured and the mean sum background pixel intensity subtracted for $N \geq 56$ cells in $n = 3$ mice.

Tetramethylrhodamine (TMRM) & Hoechst staining in live tissue *ex vivo*

Periosteum was dissected from $n = 3$ freshly killed mice, and immediately placed in warmed culture media (10% FBS in DMEM) containing 100 nM TMRM (Thermo Fisher, I34361) and 5 $\mu\text{g}/\text{mL}$ Hoechst 33342 (Thermo Fisher, H3570) for 30 minutes at 37°C. Tissue was briefly washed with fresh warmed media and imaged on a Zeiss widefield microscope (Zeiss Zen Blue Edition 2.5 software). Explanted cell controls were allowed to migrate from minced suture mesenchyme and attach for 24 hours before identical TMRM/Hoechst staining.

MC3T3 cell culture

MC3T3 cells were cultured in MEMa (Thermofisher, A1049001) 10% foetal bovine serum (FBS; Thermofisher A5256701), 1% penicillin-streptomycin (PS; Thermofisher, 15140-122). Cells were used at passages 13-20 and seeded at 25,000 cells / cm^2 for further experiments in glucose- and pyruvate-free DMEM (Thermofisher, 11966025), 10% FBS, 1% PS. All experiments were conducted in differentiation medium, consisting of glucose- and pyruvate-free DMEM 10% FBS, 1% PS, 50 μM ascorbic acid (Merck, 5960), and 2 mM glycerophosphate (Merck, 50020). Alternate serum deprivation was performed by replacing the osteogenic media by serum-free osteogenic media every 1h for 6h. For oligomycin and NMN treatments, cells were treated by 0.1 μM oligomycin (Merck, 75351), with or without 1 mM γ -nicotinamide mononucleotide (NMN; APExBio, B7878), or with 0.1% DMSO (Merck, D2438), the vehicle of oligomycin. Cells were treated for 6h before switching to differentiation medium (“oligomycin” or “DMSO” condition) or differentiation medium + 1 mM NMN (“oligomycin + NMN” condition). The media were changed twice a day.

Inhibition of ATP production in MC3T3 and measure of Reactive Oxygen Species (ROS) production

ATP level in cells was measured in 96 well plates after 6h of the treatment induction, using the CellTiter-Glo® 2.0 Assay (Promega, G9241) according to manufacturer's instructions. ROS production was measured in live cells using the CellROX deep red staining kit (Thermofisher, C10422) using a spinning disk microscope (Nikon Ti2, Crest Optics X-Light V3), with the sum projection intensity normalized by the number of cells using Hoechst staining (Fisherscientific, 11544876).

MC3T3 count and mitochondrial activity

Cell count and mitochondrial activity were measured with Hoechst and TMRM staining (Thermofisher, I34361) respectively, using a spinning disk microscope and according to manufacturer's instructions. Z-stacks of 20 µm were acquired and the TMRM sum intensity was normalized to the cell number.

Holographic imaging

Mitochondrial morphology was observed using holotomography imaging on live cells (Tomocube, HT-X1), with a reference refractive index set at 1.337. After reconstruction of the images, mitochondria were categorized in three types according to their shape. The proportion of each morphology type was determined with Tomo Analysis and Fiji softwares. MitotrackerGreen (Thermofisher, M7514) was used to stain mitochondria and confirm the mitochondrial morphology obtained by holotomography.

RT-qPCR

RNA was extracted 12h after the treatment induction (Marchery-Nagel, M36006) according to manufacturer's instructions. Reverse transcription (Thermofisher, 4374966) and qPCR (Thermofisher, 4368706) were performed according to manufacturer's instruction and mRNA target levels were normalised by beta-Actin mRNA levels using the $2^{-1\Delta\Delta C_t}$ method. The following primer have been used (5'è 3'): *MFN1*: F- AACTTGATCGAATAGCATCCGAG, R- GCATTGCATTGATGACAGAGC; *MFN2*: F- CTGGGGACCGGATCTTCTTC, R- CTGCCTCTCGAAATTCTGAAACT; *DRP1*: F- GGGCACTTAAATTGGGCTCC, R- TGTATTCTGTTGGCGTGGAAC; *PGC1α*: F- AGTGGTGTAGCGACCAATCG, R- AATGAGGGCAATCCGTCTTCA; *ARBP*: F- AGATTCGGGATATGCTGTTGG, R- AAAGCCTGGAAGAAGGAGGTC; *PINK1*: F- CACACTGTTCTCGTTATGAAGA, R- TTGAGATCCCGATGGGCAAT; *PARKIN*: F- GAGGTCCAGCAGTTAAACCCA; R- CACACTGAACTCGGAGCTTTC; *LC3II*: F- TTATAGAGCGATACAAGGGGGAG, R- CGCCGTCTGATTATCTTGATGAG; *ATG5*: F- AAGTCTGTCCTTCCGCAGTC, R- TGAAGAAAGTTATCTGGGTAGCTCA; *LONP1*: F- ATGACCGTCCCGATGTGT, R- CCTCCACGATCTTGATAAAGCG; *HSP60*: F- CACAGTCCTTCGCCAGATGAG, R- CTACACCTTGAAGCATTAAGGCT; *HPSA9*: FAATGAGAGCGCTCCTTGCTG, R- CTGTTCCCCAGTGCCAGAAC;; *MTCO1*: F- TATGTTCTATCAATGGGAGC, R-GTAGTCTGAGTAGCGTCGTG.

Evaluation of the osteoblastic differentiation

Cells were cultured for 2 weeks before the staining of the alkaline phosphatase (ALP; Abcam, ab284936) or mineralized matrix using alizarin red S (ARS; 2% solution in H₂O) (Merck, A5533). ALP staining was performed as per manufacturer's recommendation, and the stained area were quantified using Fiji on pictures taken under a light microscope. For ARS staining the cells were fixed with 4% paraformaldehyde in PBS, stained by ARS then imaged using a light microscope. The stained mineralized matrix was then dissolved with 10% (v/v) acetic acid in H₂O and optical density was measured at 405 nm to quantify the amount of dissolved ARS.

Statistical analysis

Statistical analysis was performed in Graphpad Prism 9. All data are plotted as mean, with error bars indicating standard deviation. Typically, experimental data was analyzed from all mouse conditions together using one-way ANOVA, with individual comparisons between conditions/regions of interest displayed on figures as Welch's two-tailed unpaired tests. Welch's ANOVA was used instead of ordinary ANOVA because it is more conservative and reduces the risk of false-positive p-values. Two-factors comparisons were analyzed with two-way ANOVA with uncorrected Fisher's LSD test. Only the statistical values for the hypotheses being tested are shown; multiple comparison corrections were not applied. Ratio-paired two-tailed t-test was used in comparisons within the same mouse (i.e., comparisons of Wnt3a-bandage vs Inactive-bandage). Correction of multiple comparison was not applied as each p-value was considered for itself. On some tested groups, correction for multiple comparison did not change our general conclusions. Uncorrected Dunn's tests were performed for the statistical analysis of UCell gene-expression signature scoring, as a non-parametric test for non-Gaussian distributions of data.

RNAseq sample processing

Periosteum and suture mesenchyme was separately dissected and pooled from ≥ 3 live mice from each condition, briefly minced with a scalpel blade, and immediately placed in 250 μ L of a solution of 0.5 mg/ml Liberase TL (Roche, 05401020001) and 0.1 mg/ml DNaseI (Sigma Aldrich, DN25) in Hanks' Balanced Salt Solution (HBSS, Gibco, 14170-120) and incubated for 25 minutes at 37 with shaking (600 rpm). The digestion was quenched with ice-cold 1mL buffer (3% FBS in PBS), passed through a 100 μ m cell strainer (Corning, 352360), and centrifuged (500 g, 5 minutes, 4°C). The supernatant was removed, and the cell pellet gently resuspended in staining solution containing 0.1 μ g 'hashtag' antibody, 50 μ M DRAQ5 (Invitrogen, 65-0880-92), and 1:4000 DAPI in ice-cold 3% FBS/PBS. Up to 4 samples were individually stained with TotalSeq™ antibodies 1-4 (Biolegend 155801, 155803, 155805, 155807). Samples were centrifuged, washed with 3% FBS/PBS, centrifuged again and resuspended in 3% FBS/PBS for single-cell FACS. The gating strategy took single (by forward and side scatter), nucleated (DRAQ⁺), live (DAPI⁺) cells, and up to 4 individually-hashtagged samples were multiplexed together by sorting equal cell numbers into the same collection tube, and stored on ice to preserve viability. Cells were loaded onto a 10X Chromium controllers and the library generated through single cell 10X 3' v3.1 gene expression. All libraries were sequenced together using NextSeq 2000 sequencing on a P3 flow cell.

RNAseq data processing

Initial alignment and processing were performed through the CellRanger pipeline. Further quality control, processing, and analysis was carried out through the Seurat (v4) package (Hao et al. 2021 [\[1\]](#)) and additional packages in R software (R Core Team (2022), v4.2.1). Cells were graded for proportion of mitochondrial genes, and this metric regressed. The immune cell CD45⁺ populations in the samples (Ptprc^{high}) were removed from downstream analyses. CD90⁺/CD45[−] (Thy1^{high}/Ptprc^{low}) clusters were considered the populations of interest. Differential expression of lincRNA Gm42418 was identified as an artefact and removed from the data, as has been done in other studies (Kimmel et al. 2019 [\[2\]](#)). For calculation of Gene Ontology (GO) and KEGG enrichment of sets of differentially expressed genes, ShinyGO 0.76.3 (Ge, Jung, and Yao 2020) was used to calculate false discovery rate (FDR) and fold enrichment (FE) values. Full details of differentially expressed genes in comparisons between ages, diets, and CD90⁺ subpopulations are available in Source data_RNAseq tables.

To construct the senescence signature, we obtained mouse aging-related genes from the GenAge database (<https://genomics.senescence.info/genes/search.php?organism=Mus+musculus&show=4> [\[3\]](#)). The genes were categorized into two groups: positively regulated and negatively regulated. Positively regulated genes are defined as those with high expression levels associated with

senescence, while negatively regulated genes are defined as those with low expression levels associated with senescence. The genes were clustered based on their expression, then we selected the highly expressed cluster in the positively-regulated genes and the poorly expressed cluster in negatively-regulated genes in the Aged group.

For the glycolysis and lipid catabolic processes signatures, we obtained mouse metabolism-related genes from AmiGO2 database (<https://amigo.geneontology.org/amigo>) with Ontology ID: GO:0061621 (glycolysis) and GO:0050996 (positive regulation of lipid catabolic process), and we clustered the genes based on their expression levels. Heatmaps were drawn by R package pheatmap (version 1.0.12).

Data availability

Upon revision/publication all numerical data included in the production of figures, plots and statistical analysis will be found in the figures, main text, and Source Data. Additional statistical analysis values (F values, t values, degrees of freedom and p values) will be presented in the Source Data documents. Source data will be provided with this paper. Raw and processed single cell RNAseq data will be available under a GEO accession number during the revision/review process. Code used in the processing of RNAseq data will be made available upon request.

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Additional information

Author Contributions

S.J.H. conceived the project and with J.R. designed the experiments. J.R. performed surgeries, *ex vivo* experiments, *in vitro* experiments, and collected all data. J.R. analyzed microCT data, and live *ex vivo* staining

J.R and P.B analyzed histological staining. P.T and P.B performed experiments in Supplementary Figure 10 and 11. R.C. processed the metagenomic data. J.R., Y.T, A.V. and D.F. performed RNAseq analysis. J.R., P.T. and P.B. prepared the figures. J.R. and S.J.H. interpreted the data and wrote the initial manuscript draft. S.J.H revised the manuscript. All authors approved the manuscript. S.J.H. led the project and provided financial support for the project.

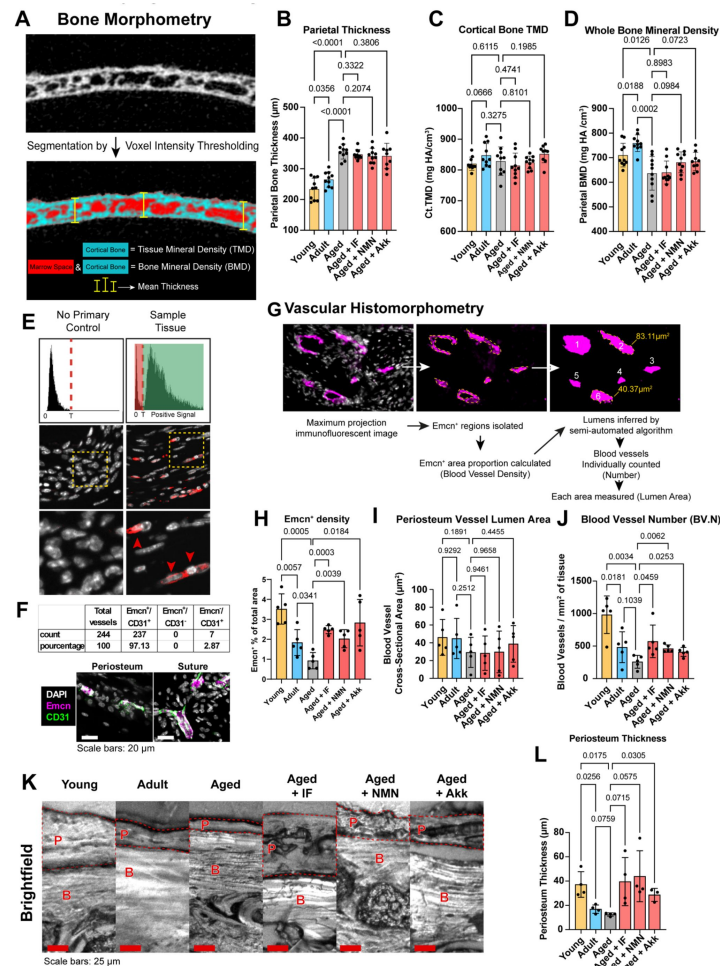
Competing interests

S.J. H is listed as an inventor on a patent application (P6125GB00) on Tissue Regeneration Patch that includes the Wnt-bandage.

Supplementary Information

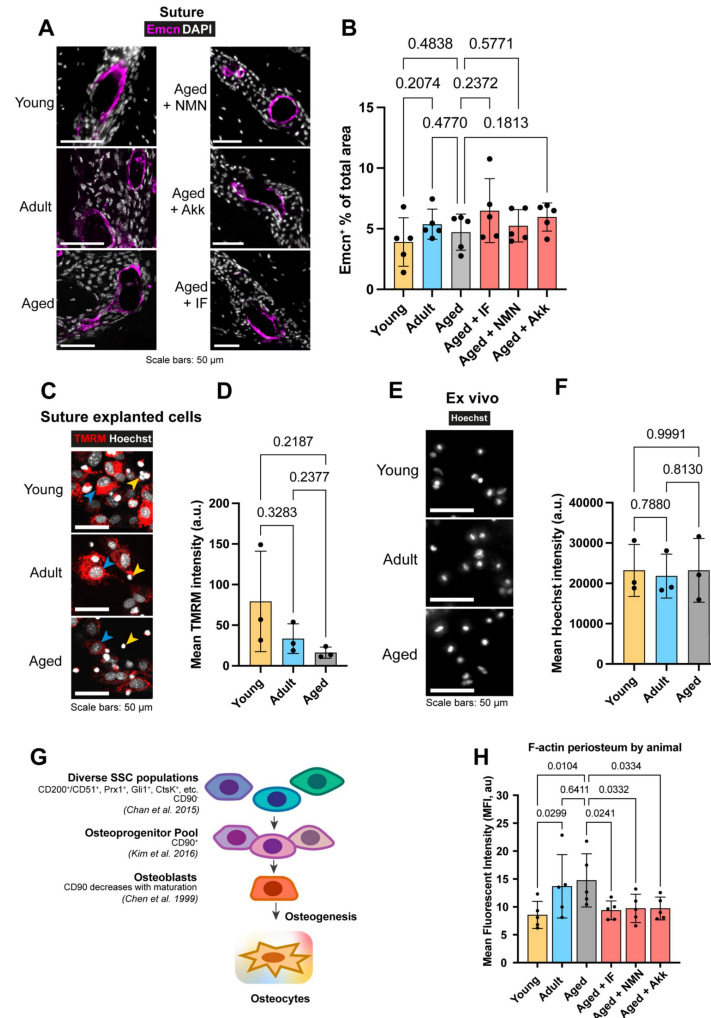
Supplementary Materials and Methods TRAP activity staining

Sections were rehydrated with ddH₂O and then processed for TRAP activity staining using a commercial kit (Takara, MK300), according to manufacturer's instruction. Briefly, sections were fixed for 5 min using the fixation solution from the kit, washed 3 times with sterile distilled water and then incubated for 45 min at 37 °C with the ACP substrate containing sodium tartrate. Sections were then washed 3 times with sterile distilled water, counterstained using Mayer's hematoxylin (Sigma Aldrich, MHS) for 1 min, washed again 3 times with tap water before mounting using glycerol. Tissue sections were imaged using a Zeiss Axioscan Z1 slide scanner and analyzed through the Zen software. The number of TRAP positive multinuclear cells normalized per bone tissue surface was reported.



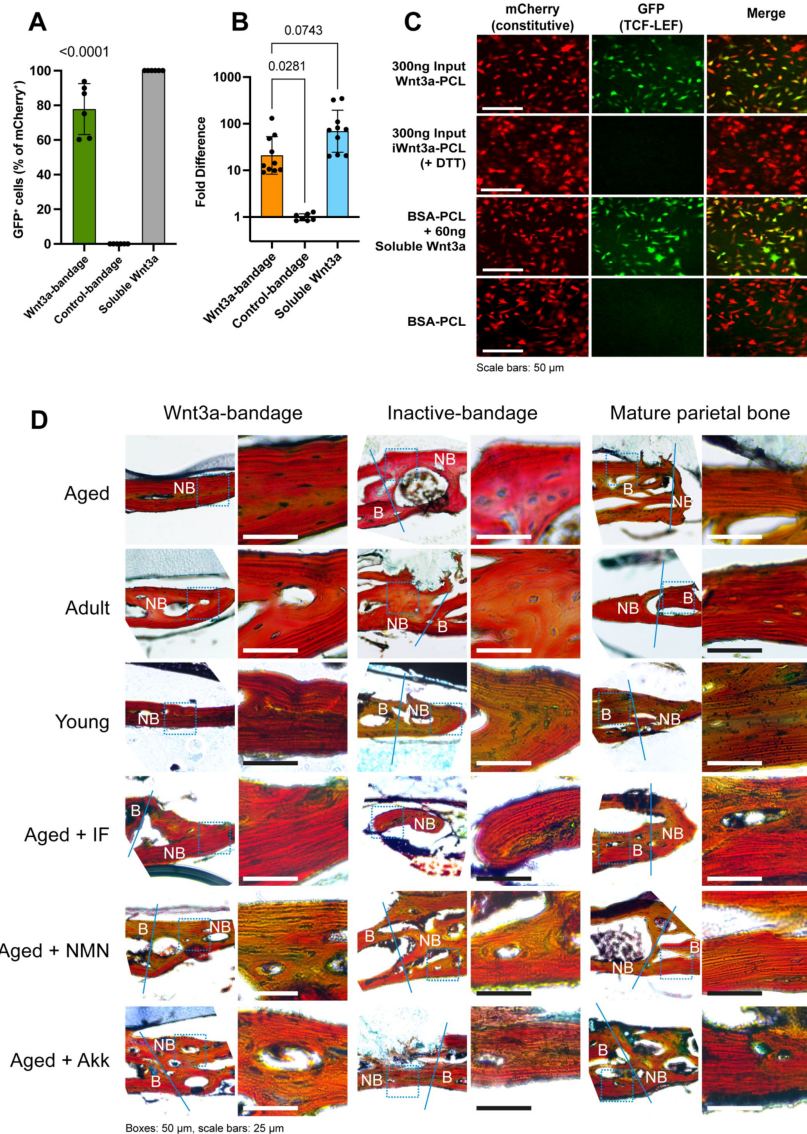
Supplementary Figure 1

: (A) Methodology for segmentation of micro-computed tomography (microCT) images for calculation of bone morphometry metrics: (B) Parietal Thickness (C) Cortical Tissue Mineral Density (Ct.TMD), (D) Parietal Whole Bone Mineral Density (BMD). Statistical analysis: Welch's tests, $n = 10$ mice per condition, except Aged + Akk ($n = 9$). (E) The method for quantifying immunofluorescence-positive (red) cells. The maximum pixel intensity is determined in control tissues lacking primary antibody (no primary control). This value is used to threshold the stained sample tissue, with the remaining signal distinguishing positive cells (red arrows). DAPI (grey) is also shown for reference of cellular locations. (F) Number and percentage of blood vessels positive for Emcn and/or CD31 with representative images in the periosteum and suture mesenchyme. (G) Methodology for segmentation of Endomucin (Emcn) immunofluorescent staining in fixed tissue sections, for calculation of vascular histomorphometry metrics: (H) Emcn⁺ staining density (proportion of total tissue area). Data are shown previously in main Figures 1, 3, and 5, and are reproduced here for reference, (I) Mean lumen area per blood vessel, (J) Number of blood vessels per area tissue (BV.N). Statistical analysis: Welch's tests, $n = 5$ mice per condition. (K) Representative brightfield images of unstained tissue sections highlighting the overlying periosteum (red P, red dashed line) and parietal bone (red B). (L) Mean periosteal thickness. Statistical analysis: Welch's tests, $n = 4$ mice per condition, except Aged + Akk ($n = 3$).



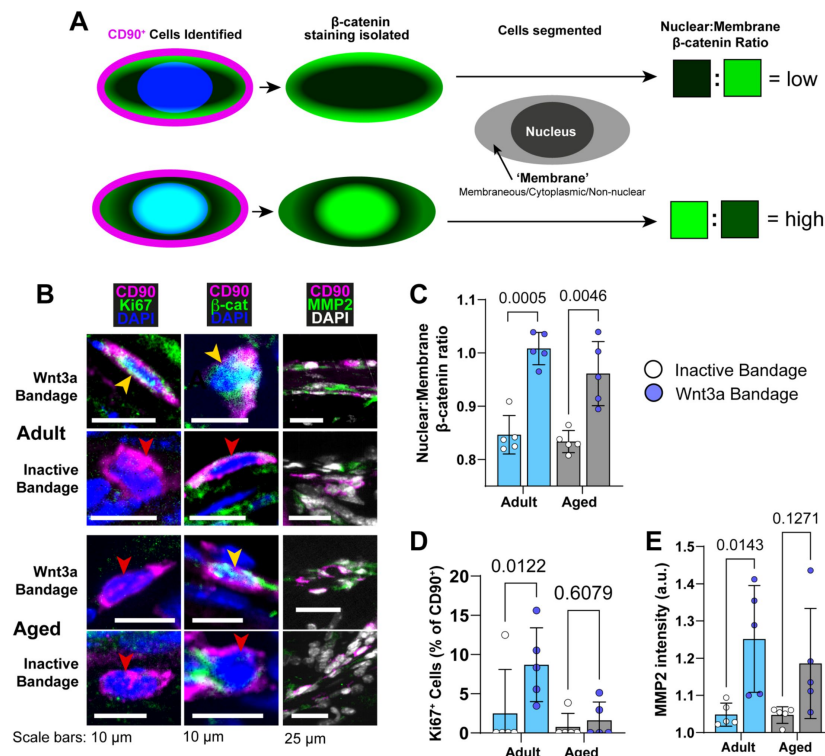
Supplementary Figure 2

: (A) representative immunofluorescence staining of Emcn (magenta) and DAPI (grey) and (B) mean cross sectional area of Endomucin (Emcn)-positive vascular cells in mouse sutures. Statistical analysis: Welch's tests in $n = 5$ mice. (C) Representative images and (D) mean intensity of live TMRM staining (red) and Hoechst (grey) in adherent cells allowed to explant from suture mesenchyme tissue for 24 hours *in vitro*. Remaining mesenchyme tissue was removed prior to staining. Statistical analysis: Welch's tests in tissue from $n = 3$ mice per condition. (E) Representative images and (F) mean intensity of live Hoechst staining (grey) from the TMRM experiment in Fig. 1i-j. Statistical analysis: Welch's tests in tissue from $n = 3$ mice per condition. (G) Schematic demonstrating the hierarchy and selected markers of Skeletal Stem Cells (SSCs), leading to the CD90⁺ cell pool, which ultimately differentiates into osteoblasts and osteocytes during osteogenesis. (H) Mean Phalloidin-488 staining intensity for F-actin in the periosteum. Statistical analysis: Welch's tests in $n = 5$ mice.



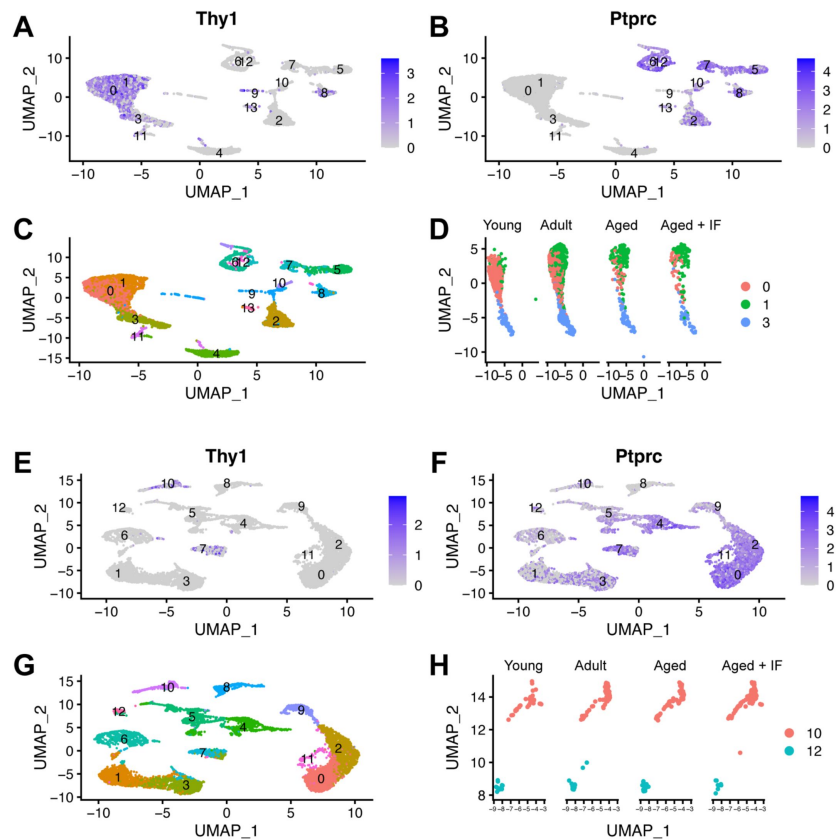
Supplementary Figure 3

: **(A)** Characterization of Wnt3a-bandage activation of the Wnt/β-catenin pathway with Comma-D Beta cells *in vitro*. Presented as the proportion of GFP⁺ cells, normalized to the proportion of GFP⁺ responsive to soluble Wnt3a. Statistical analysis: two-tailed one-sample t test against the inactive control-bandage (mean = 0); n = 6 for each condition. **(B)** Characterization of Wnt3a-bandage activation of the Wnt/β-catenin pathway with LS/L cells *in vitro*. Presented as fold change of Luciferase signal over control-bandage. Statistical analysis: Welch's test, n = 10 for each Wnt3a-bandage and Soluble Wnt3a, n = 7 for Control-bandage. **(C)** Representative images for Comma-D Beta cells quantified in **(A)**. mCherry (red) is constitutively expressed, GFP (green) is expressed during Wnt/β-catenin pathway activation. Wnt3a is treated with DTT, resulting in complete chemical inactivation (iWnt3a). Bovine Serum Albumin (BSA) is used as an additional negative control. Polycaprolactone (PCL) is the polymeric material that Wnt3a is bound to during Wnt3a-bandage manufacture. **(D)** Representative Movat's pentachrome staining of bone tissues showing the lamellar/striated structures. Blue line designates the drilled defect edge, thereby separating original intact bone B from new bone NB. The dashed blue box shows the enlarged region.



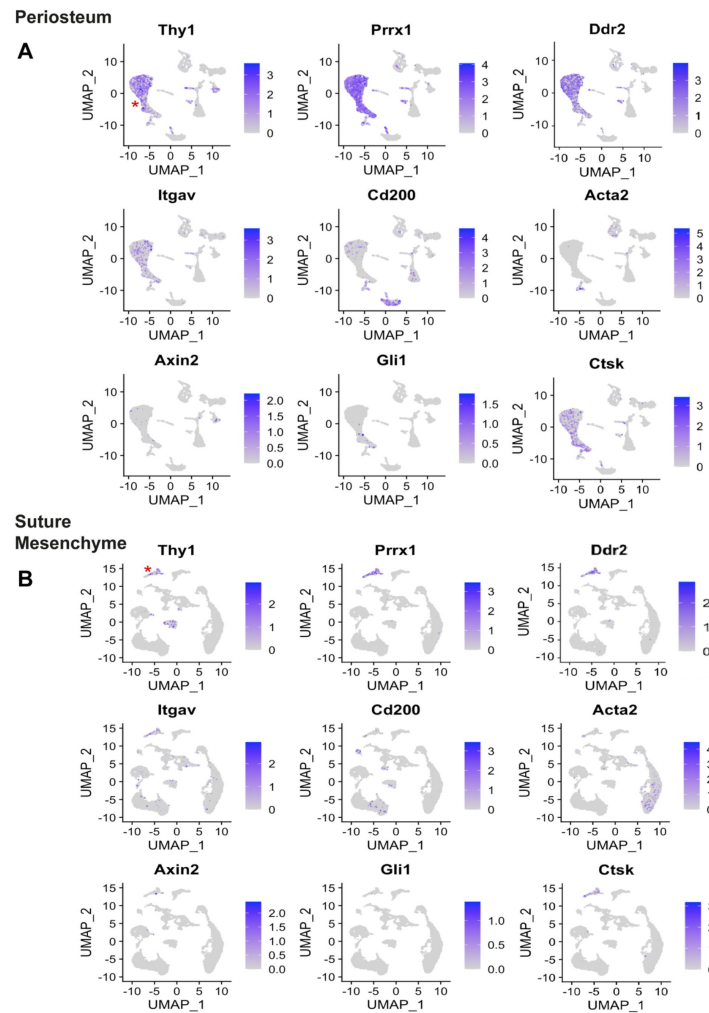
Supplementary Figure 4

: (A) Schematic depicting the methodology for measuring the ratio of Nuclear:Membrane β -catenin in single CD90⁺ cells. (B) Representative images depicting CD90 (magenta), DAPI (grey/blue as indicated by column), and functional markers (green) as indicated by column. Yellow arrows indicate marker-high/positive cells; red arrows indicate marker-low/negative cells. (C) Proportion of CD90⁺ cells that are Ki67⁺ by age. Statistical testing: paired, two-tailed t tests in n = 5 mice per condition. (D) Ratio of Nuclear:Membrane β -catenin staining in CD90⁺ cells by age. Statistical testing: ratio paired, two-tailed t tests in n = 5 mice per condition. (E) Normalized intensity staining for MMP2 in CD90⁺ cells by age. Statistical testing: ratio paired, two-tailed t tests in n = 5 mice per condition. For all plots (C-E), Wnt3a-bandage (dark blue points), inactive bandage (white points), Adult (blue), Aged (grey).



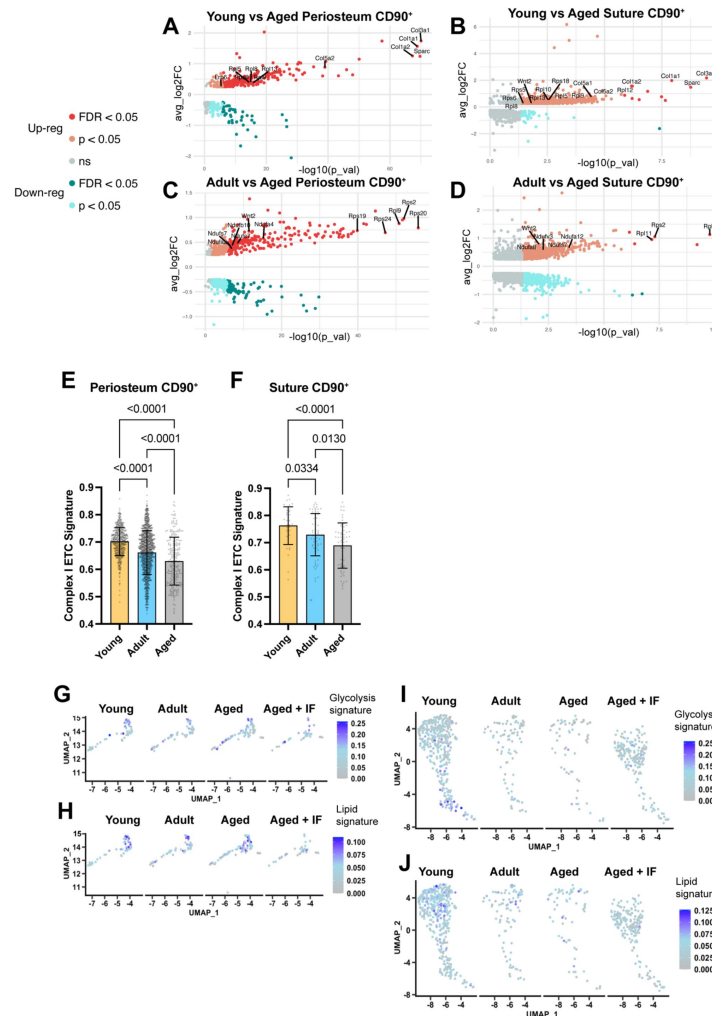
Supplementary Figure 5

: UMAP-coordinate plot of Thy1 (CD90, **A**) and Ptprc (CD45, **B**) expression in clusters of cells within the periosteum. (**C**) Periosteum cell populations (clusters) as identified through unbiased clustering (**D**) The CD90⁺/CD45⁻ clusters of interest (0, 1, 3), separated by mouse condition. UMAP-coordinate plot of Thy1 (CD90, **E**) and Ptprc (CD45, **F**) expression in clusters of cells within the suture. (**G**) Suture mesenchyme cell populations (clusters) as identified through unbiased clustering (**H**) The CD90⁺/CD45⁻ cluster of interest (10), and also the small endothelial cluster (12), separated by mouse condition.



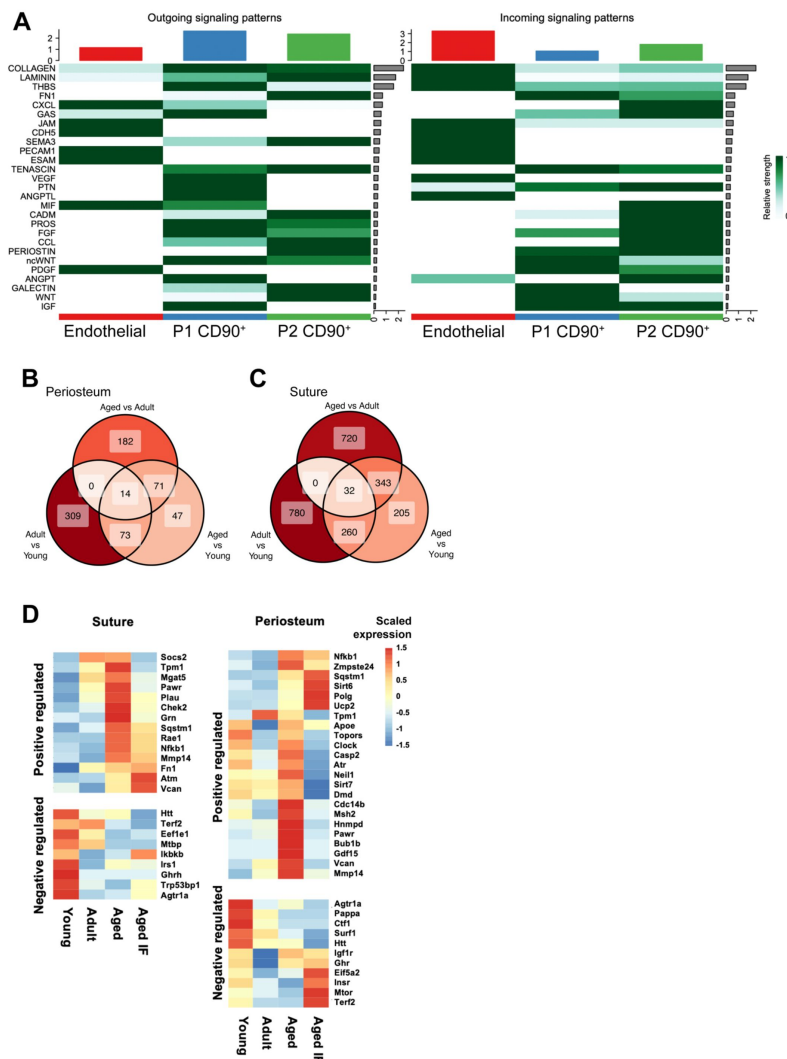
Supplementary Figure 6

: UMAP-coordinate plot of the entire cell populations from Periosteum (**A**) and Suture Mesenchyme (**B**). Expression of SSC markers (purple) highlights clusters of transcriptomically-similar cells. The red asterisk is adjacent to the group of cells referred to as the CD90⁺ (Thy1) population, which is also CD45⁺ (Ptprc⁺) (**Fig. 5a-h**). Prrx1 (Prx1), Ddr2, Itgav (CD51), Cd200 (CD200), Acta2(a-SMA), Axin2, Gli1 and Ctsk (Cathepsin K) are shown.



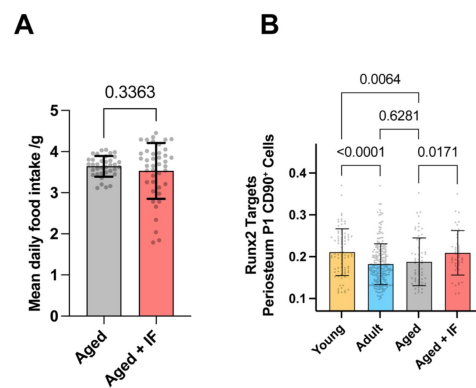
Supplementary Figure 7

: (A) Volcano plot showing differentially-expressed genes in the comparison of the CD90⁺ population in Young vs Aged periosteum; (B) Young vs Aged suture; (C) Adult vs Aged periosteum and (D) Adult vs Aged Suture. Statistical analysis: Wilcoxon Rank Sum test via the FindMarkers function in the Seurat R package. Upregulated (pink: $p < 0.05$; red: FDR-adjusted $p < 0.05$), downregulated (cyan: $p < 0.05$; teal: FDR-adjusted $p < 0.05$), and non-significant (ns) genes are displayed on axes of log fold expression (y, avg_log2FC) against p transformed as its negative log value. (E) UCell Signature Scoring of transcript expression of genes relating to Complex I electron transport chain (GO-term: 0005747) in the CD90⁺ clusters of the periosteum and (F) suture. Statistical analysis: Uncorrected Dunn's tests; $n \geq 242$ cells in periosteum per age group, $n \geq 39$ cells in suture per age group. (G-J) UMAP-coordinate plot of the CD90⁺ cells in the periosteum (G, H) and suture mesenchyme (I, J) for the glycolysis and lipid UCell Signature scoring.



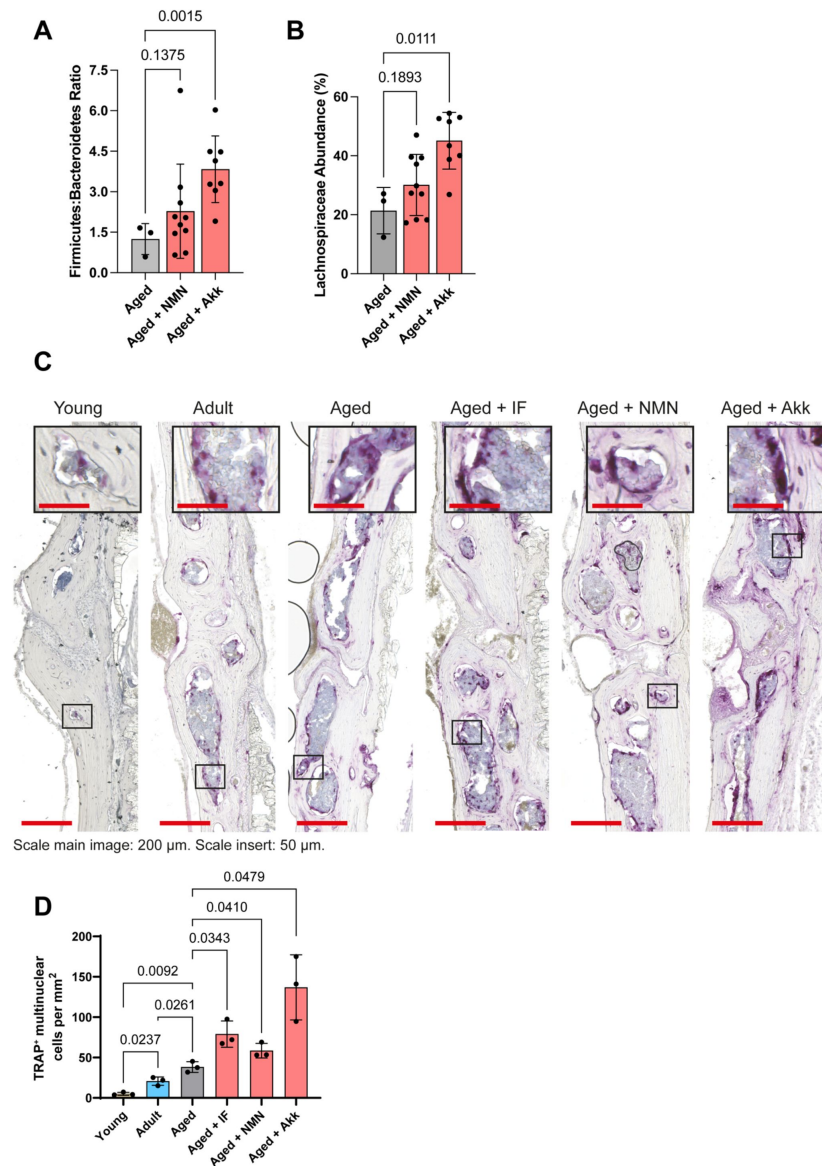
Supplementary Figure 8

(A) CellChat analysis of the relative inferred levels of outgoing and incoming signaling between subsets of periosteal cells, grouped into pathways. P2 contains the Igf1^{high} and Igf1^{low} populations. Green color scale shows relative intensity in outgoing (left) and incoming (right) inferred signaling patterns for pathways shown at the left edge. Red/blue/green boxes at the top show the relative sum of all pathways. (B) Three-way Venn diagrams showing the overlap of statistically-significant positively-upregulated differentially-expressed genes ($p < 0.05$) in comparisons between the CD90⁺ clusters in the periosteum and (C) suture of Young, Adult, and Aged mice. Color of shading and label shows number of genes shared. The central section shows the number of genes conserved throughout all comparisons. (D) Heatmaps of gene expression in CD90⁺ cells from suture and periosteum mesenchyme of young, adult, aged and aged + IF mice. Scaled expression of genes positively regulated and negatively regulated in senescent cells.



Supplementary Figure 9

: **(A)** The mean daily food intake of Aged mice on ad libitum (AL) diet vs intermittent fasting (IF), averaged over the course of measurements of $n \geq 10$ mice. Statistical analysis: Welch's t-test of $n \geq 36$ total food measurements taken. **(B)** UCell Signature Scoring of transcript expression of transcriptional targets of Runx2 in the P1 CD90⁺ cluster subset of the periosteum. Statistical analysis: Uncorrected Dunn's tests; $n \geq 46$ cells per condition.

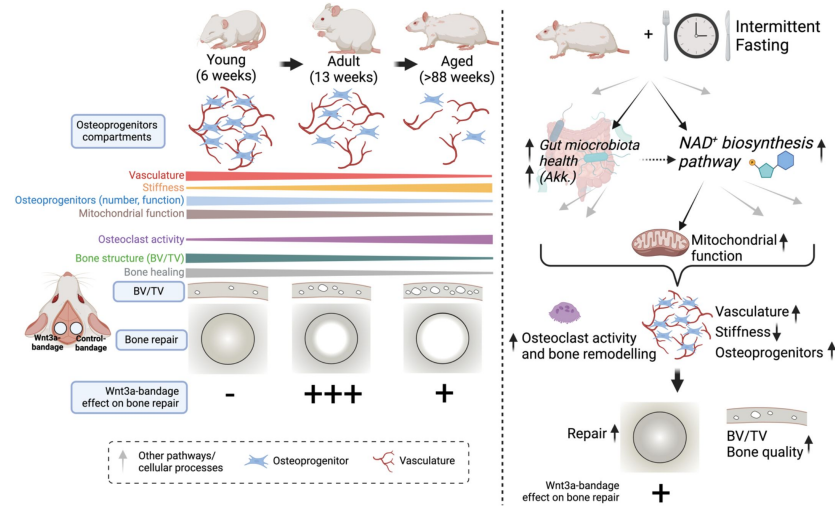


Supplementary Figure 10

: (A) The relative abundance of Firmicutes to Bacteroidetes bacteria and (B) Lachnospiraceae bacteria, as measured by 16S metagenomic profiling. Statistical analysis: Welch's tests on $n = 3$ (Aged), $n = 10$ (Aged + NMN), $n = 8$ (Aged + Akk). (C) Representative calvarial TRAP activity staining images. (D) Number of TRAP positive multinuclear cells normalized per bone tissue surface. Statistical analysis: Welch's tests $n = 3$ mice per group. The black box shows the enlarged region.

Supplementary Figure 11

: Schematic that recapitulates the mains findings of the study.



Supplementary Table 1

: Statistical analysis of Fig. 1a. Stars and symbols indicate statistical significance calculated by one-way ANOVA followed by unpaired t-test with Welch's correction, or by ratio paired t-test (non-significant, $p > 0.05$) when comparing inactive vs active within the same group (e.g. Young inactive vs Young active). Ns: non significant, * $p < 0.05$, ** $p < 0.01$,

		Inactive						Active					
		Young	Adult	Aged	Aged + IF	Aged + NMN	Aged + Akk	Young	Adult	Aged	Aged + IF	Aged + NMN	Aged + Akk
Inactive	Young	*											
	Adult	****	*										
	Aged			****									
	Aged + IF	*	ns	****	ns								
	Aged + NMN	*	ns	****	*								
Active	Young	ns	*	****	*	ns	***						
	Adult	ns	*	****	*	*	***	ns					
	Aged	****	*	*	***	**	*	****	****				
	Aged + IF	ns	ns	****	ns	ns	**	ns	ns	***			
	Aged + NMN	ns	ns	****	ns	ns	**	ns	ns	***	ns		
	Aged + Akk	ns	ns	****	ns	ns	**	ns	ns	****	ns	ns	

Supplementary Table 2

: RNAseq comparison data summary

Comparison	Tissue	Number of genes significantly impacted	Effect	Gene or pathway impacted (GO term)	Figure
CD90 ⁺ cells Young vs Aged	Periosteum	59	upregulated in Young	Cytosolic ribosome (GO: 0022626), FDR: 1.0E-30, FE: 14.2	Supplementary Fig. 7a
	Suture mesenchyme	879		Collagen-containing extracellular matrix (GO: 0062023), FDR: 2.0E-29, FE: 6.8	Supplementary Fig. 7a
CD90 ⁺ cells Young vs Adult	Periosteum	1	down regulated in Adult	Structural constituent of ribosome (GO: 0003735), FDR: 2.45E-3, FE: 8.15	Supplementary Fig. 7b
CD90 ⁺ cells Young vs Aged	Suture mesenchyme	1	down regulated in Aged	Extracellular matrix structural constituent (GO: 0005201), FDR: 7.65E-08, FE: 4.58	Supplementary Fig. 7b
CD90 ⁺ cells Adult vs Aged	Suture mesenchyme	1	down regulated in Aged	Wnt2 transcripts downregulated (FDR adjusted p value (q) = 1.16E-07)	Supplementary Fig. 7c
CD90 ⁺ cells Young vs Adult	Periosteum	1	down regulated in Adult	Wnt2 transcripts downregulated (FDR adjusted p value (q) = 1.01E-09)	RNA-seq Table 5
CD90 ⁺ cells Young vs Aged	Periosteum	1	down regulated in Adult	Wnt2 transcripts downregulated (non-adjusted p value (p) = 1.58E-02)	Supplementary Fig. 7b
CD90 ⁺ cells Young vs Adult	Periosteum	1	down regulated in Adult	Wnt2 transcripts downregulated (p = 8.58E-03)	Supplementary Fig. 7d
CD90 ⁺ cells Young vs Aged	Periosteum	1	down regulated in Aged	Lrp6 declined Adult vs Young (p = 4.14E-18)	Supplementary Table 5
CD90 ⁺ cells Adult vs Aged	Periosteum	637	upregulated in Adult	Lrp6 declined Aged vs Young (p = 4.01E-06)	Supplementary Fig. 7a
	Suture mesenchyme	610		NADH dehydrogenase complex assembly (GO: 0010257), FDR: 9.9E-16, FE: 12.8	(Supplementary Fig. 7c)
CD90 ⁺ cells Adult vs Aged	Periosteum		downregulated in Aged	NADH dehydrogenase (ubiquinone) activity (GO: 0008137), FDR: 7.1E-09, FE: 14.4	(Supplementary Fig. 7c)
CD90 ⁺ cells Adult vs Aged	Periosteum		upregulated in Igf1 low	Oxidative phosphorylation (GO: 0006119), FDR: 7.2E-20, FE: 9.6	(Supplementary Fig. 7c)
CD90 ⁺ cells Adult vs Aged	Periosteum		downregulated in Aged	Mitochondrial ATP synthesis coupled proton transport (GO: 0042776), FDR: 2.7E-09, FE: 9.3	Supplementary Fig. 7d
CD90 ⁺ cells Adult vs Aged	Periosteum		upregulated in Adult	Oxidative phosphorylation (GO: 0006119), FDR: 5.6E-16, FE: 8.7	Supplementary Fig. 7d
CD90 ⁺ cells Adult vs Aged	Periosteum		downregulated in Aged	Complex I gene (GO: 0005747) (p < 0.0001)	Supplementary Fig. 9b-c
CD90 ⁺ cells Adult vs Aged	Periosteum	85	upregulation in Igf1 low	Cell motility (GO: 0048870), FDR: 2.58E-08, FE: 4.40	Fig. 2c
CD90 ⁺ cells Igf1 low vs Igf1 high	Periosteum	204	upregulation in Igf1 high	Collagen-containing extracellular matrix (GO: 0062023), FDR: 6.18E-38, FE: 14.41	Fig. 2c
CD90 ⁺ cells proliferating vs non proliferating	Suture mesenchyme	855	upregulation in proliferating cells	Oxidative phosphorylation (KEGG: mmu00190, FDR: 2.08E-03, FE: 5.97)	Fig. 2c
	Periosteum	98		Respiratory chain complex (GO: 0098803), FDR: 1.43E-03, FE: 8.61	Fig. 2c
CD90 ⁺ cells Aged vs Aged IF	Suture mesenchyme	297	upregulation in Aged IF	Mitotic cell cycle (GO: 0000278), FDR: 1.0E-47, FE: 4.5	Fig. 2d-e
	Periosteum	3		ATP-dependent activity (GO: 0140657), FDR: 4.8E-08, FE: 2.6	Fig. 2d-e
CD90 ⁺ cells Aged vs Aged IF	Suture mesenchyme	1	upregulation in Aged IF	Response to cytokine (GO: 0034097), FDR: 2.91E-08, FE: 5.63	Fig. 4a
	Suture mesenchyme	>5		Mitotic cell cycle (GO: 0000278), FDR: 1.2E-05, FE: 3.2	Fig. 4b
CD90 ⁺ cells Aged vs Aged IF	Suture mesenchyme	7	upregulation in Aged IF	Protein phosphorylation (GO: 0006468), FDR: 1.7E-05, FE: 2.4	Fig. 4b
	Suture mesenchyme	7		Dbp, Hlf, and Tef (each q < 0.05)	Supplementary Fig. 7e-f
CD90 ⁺ cells Aged vs Aged IF	Suture mesenchyme	7	upregulation in Aged IF	Dbp, Hlf, and Tef (each q < 0.05)	Supplementary Fig. 7e-f
	Suture mesenchyme	7		Genes under circadian control (e.g., Sirt1, Kat2b, Csnk1e, Ezh2, Fbxw11, Ucp2; all p < 0.05)	Fig. 4b
CD90 ⁺ cells Aged vs Aged IF	Suture mesenchyme	7	upregulation in Aged IF	transcriptional targets of Clock/Per2/Amf (FDR: 6.42E-04, 2.95E-03, 1.23E-02 respectively)	RNA-seq Table 12
	Suture mesenchyme	7			

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Reviewer #1 (Public review):

Summary:

Aging reduces tissue regeneration capacity, posing challenges for an aging population. In this study, the authors investigate impaired bone healing in aging, focusing on calvarial bones, and introduce a two-part rejuvenation strategy. Aging depletes osteoprogenitor cells and reduces their function, which hinders bone repair. Simply increasing the number of these cells does not restore their regenerative capacity in aged mice, highlighting intrinsic cellular deficits. The authors' strategy combines Wnt-mediated osteoprogenitor expansion with intermittent fasting, which remarkably restores bone healing. Intermittent fasting enhances osteoprogenitor function by targeting NAD⁺ pathways and gut microbiota, addressing mitochondrial dysfunction - an essential factor in aging. This approach shows promise for rejuvenating tissue repair, not only in bones but potentially across other tissues.

Strengths:

This study is exciting, impressive, and novel. The data presented is robust and supports the findings well.

Weaknesses:

As mentioned above the data is robust and supports the findings well. I have minor comments only.

<https://doi.org/10.7554/eLife.104068.1.sa2>

Reviewer #2 (Public review):

Summary:

Reeves et al explore a model of bone healing in the context of aging. They show that intermittent fasting can improve bone healing, even in aged animals. Their study combines a 'bone bandage' which delivers a canonical Wnt signal with intermittent fasting and shows impacts on the CD90 progenitor cell population and the healing of a critical-sized defect in the calvarium. They also explore potential regulators of this process and identify mitochondrial dysfunction in the age-related decline of stem cells. In this context, by modulating NAD⁺ pathways or the gut microbiota, they can also enhance healing, hinting at an effect mediated by complex impacts on multiple pathways associated with cellular metabolism.

Strengths:

The study shows a remarkable finding: that age-related decreases in bone healing can be restored by intermittent fasting. There is ample evidence that intermittent fasting can delay aging, but here the authors provide evidence that in an already-aged animal, intermittent

fasting can restore healing to levels seen in younger animals. This is an important finding as it may hint at the potential benefits of intermittent fasting in tissue repair.

Weaknesses:

The authors explore potential mechanisms by which the intermittent fasting protocol might impact bone healing. However, they do not identify a magic bullet here that controls this effect. Indeed, the fact that their results with intermittent fasting can be replicated by changing the gut microbiota or modulating fundamental pathways associated with NAD, suggests that there is no single mechanism that drives this effect, but rather an overall complex impact on metabolic processes, which may be very difficult to untangle.

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Reviewer #3 (Public review):

Summary:

This study aims to address the significant challenge of age-related decline in bone healing by developing a dual therapeutic strategy that rejuvenates osteogenic function in aged calvarial bone tissue. Specifically, the authors investigate the efficacy of combining local Wnt3a-mediated osteoprogenitor stimulation with systemic intermittent fasting (IF) to restore bone repair capacity in aged mice. The highlights are:

(1) Novel Approach with Aged Models:

This pioneering study is among the first to demonstrate the rejuvenation of osteoblasts in significantly aged animals through intermittent fasting, showcasing a new avenue for regenerative therapies.

(2) Rejuvenation Potential in Aged Tissues:

The findings reveal that even aged tissues retain the capacity for rejuvenation, highlighting the potential for targeted interventions to restore youthful cellular function.

(3) Enhanced Vascular Health:

The study also shows that vascular structure and function can be significantly improved in aged tissues, further supporting tissue regeneration and overall health.

Through this innovative approach, the authors seek to overcome intrinsic cellular deficits and environmental changes within aged osteogenic compartments, ultimately achieving bone healing levels comparable to those seen in young mice.

Strengths:

The study is a strong example of translational research, employing robust methodologies across molecular, cellular, and tissue-level analyses. The authors leverage a clinically relevant, immunocompetent mouse model and apply advanced histological, transcriptomic, and functional assays to characterise age-related changes in bone structure and function. Major strengths include the use of single-cell RNA sequencing (scRNA-seq) to profile osteoprogenitor populations within the calvarial periosteum and suture mesenchyme, as well as quantitative assessments of mitochondrial health, vascular density, and osteogenic function. Another important point is the use of very old animals (up to 88 weeks, almost 2 years) modelling the human bone aging that usually starts >65 yo. This comprehensive approach enables the authors to identify critical age-related deficits in osteoprogenitor number, function, and microenvironment, thereby justifying the combined Wnt3a and IF intervention.

Weaknesses:

One limitation is the use of female subjects only and the limited exploration of immune cell involvement in bone healing. Given the known role of the immune system in tissue repair, future studies including a deeper examination of immune cell dynamics within aged osteogenic compartments could provide further insights into the mechanisms of action of IF.

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